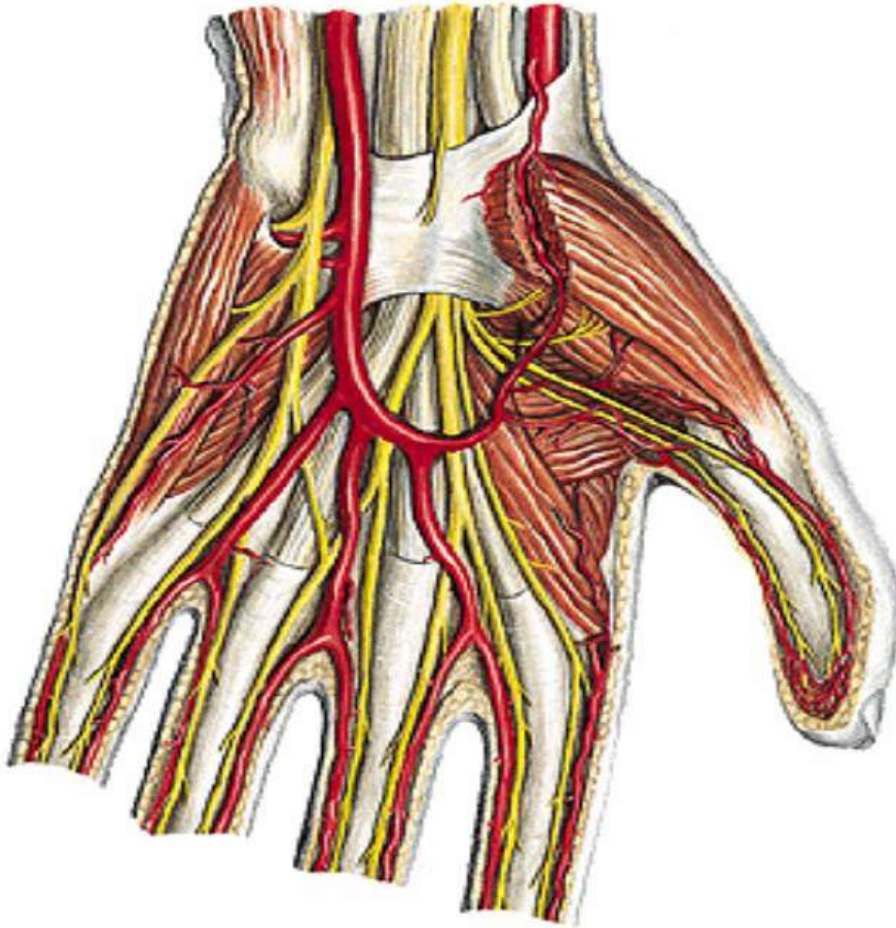


THE HAND



Prof. Laila M. Aboul Mahasen

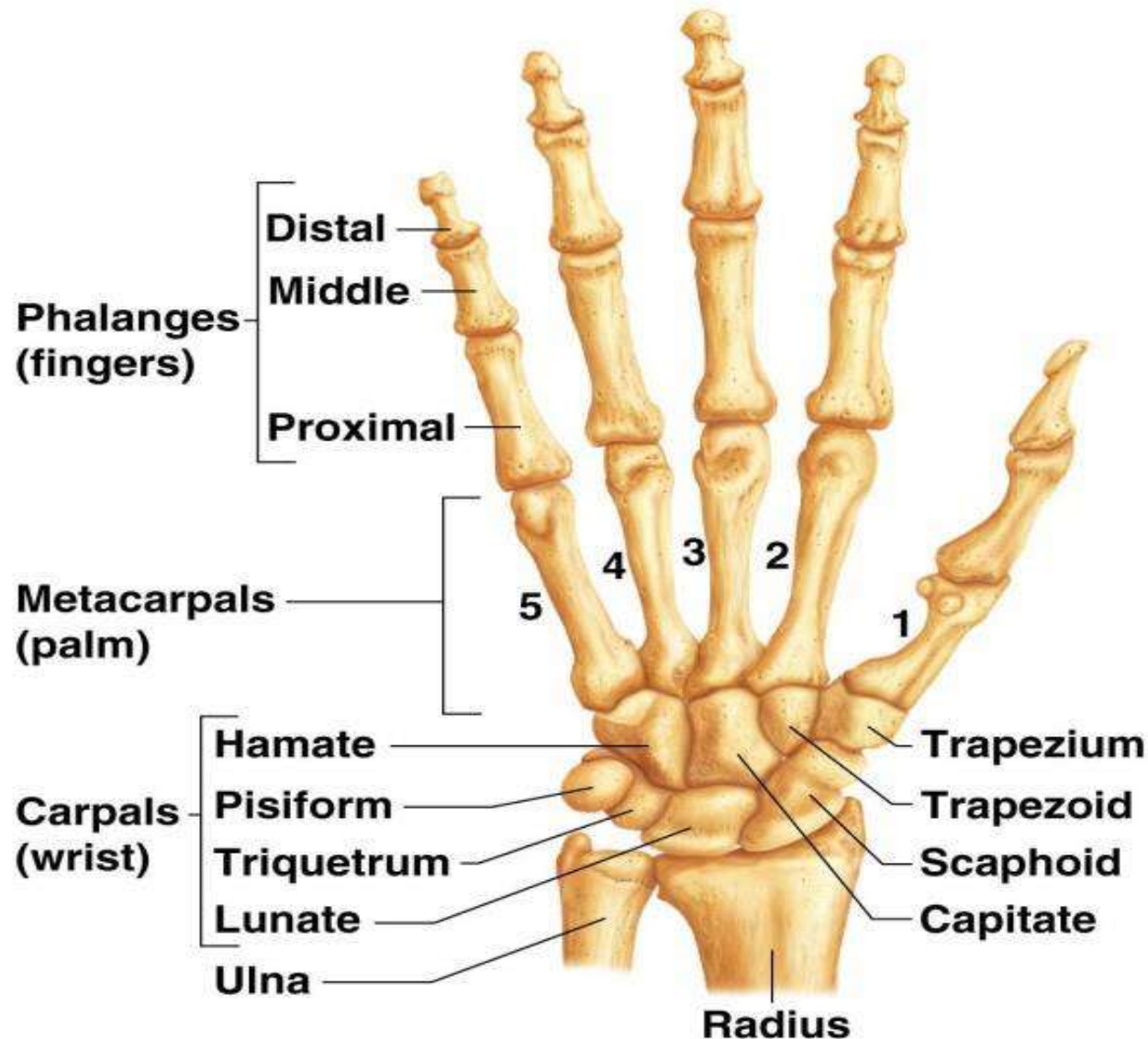
Anatomy Dept. , Faculty of Medicine , KAU

Objectives

By the end of this lecture, you should be able to:

1. Develop clear concept of the bones, muscles, vessels, and nerves of the hand.
2. Demonstrate the cutaneous innervation of the hand.
3. Enumerate the short muscles of the hand (names, actions, and nerve supply).
4. Understand the formation of superficial and deep palmar arches.

Bones of the hand



Bones of the hand

The bones of the hand and wrist are arranged into three main types :

1. The carpals (8 forming the wrist bones)
2. The metacarpals (5 forming the palm bones)
3. The phalanges (14 forming fingers bones).

Bones of the wrist (carpals) : are arranged in **two rows** from lateral to medial :

Proximal row (closer to the forearm) : **Scaphoid, Lunate, Triquetrum and Pisiform**

Distal row (closer to the palm): **Trapezium, Trapezoid, Capitate and Hamate**

N.B.(She Looks Too Pretty –Try To catch Her)

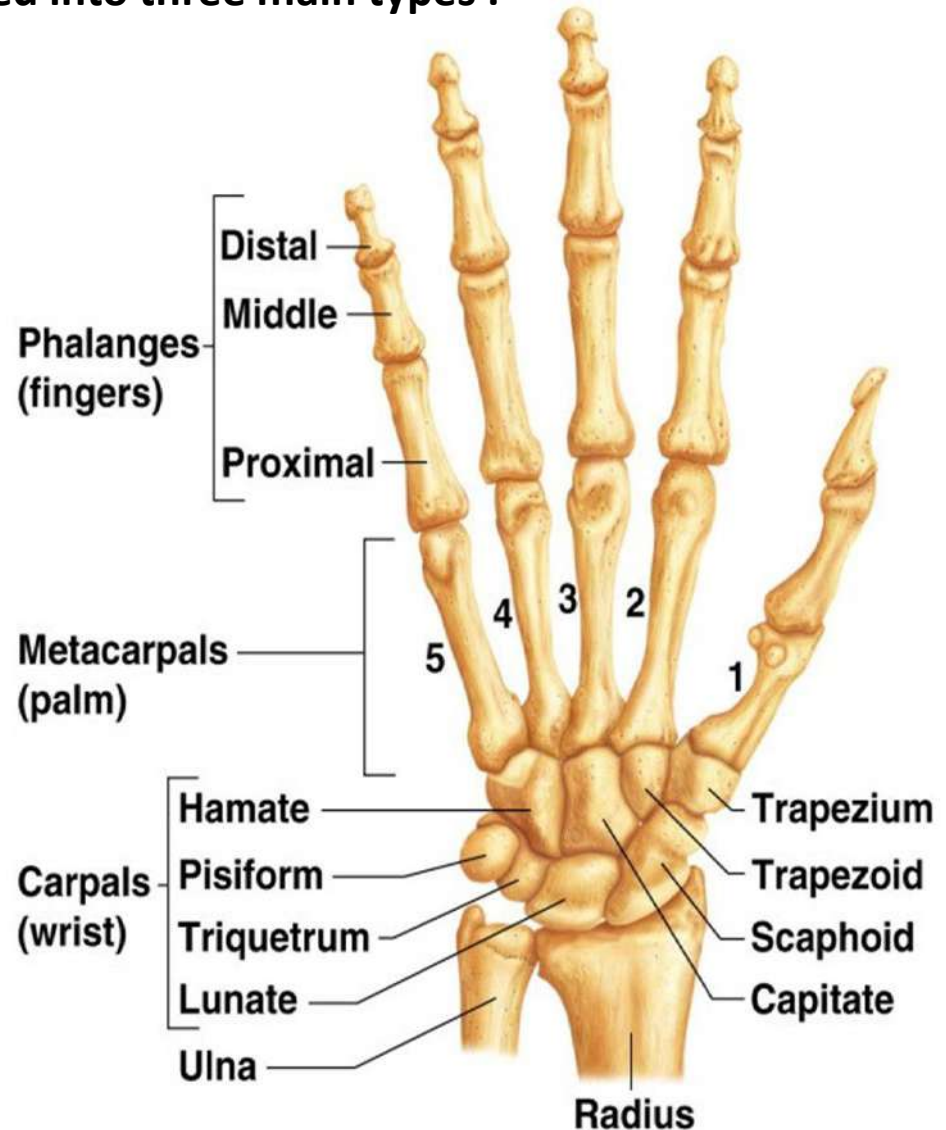
Bones of the palm (5 Metacarpal bones) :
(each one has base, head & shaft)

-The 1st metacarpal is that of thumb . It is the is shortest & most mobile one .

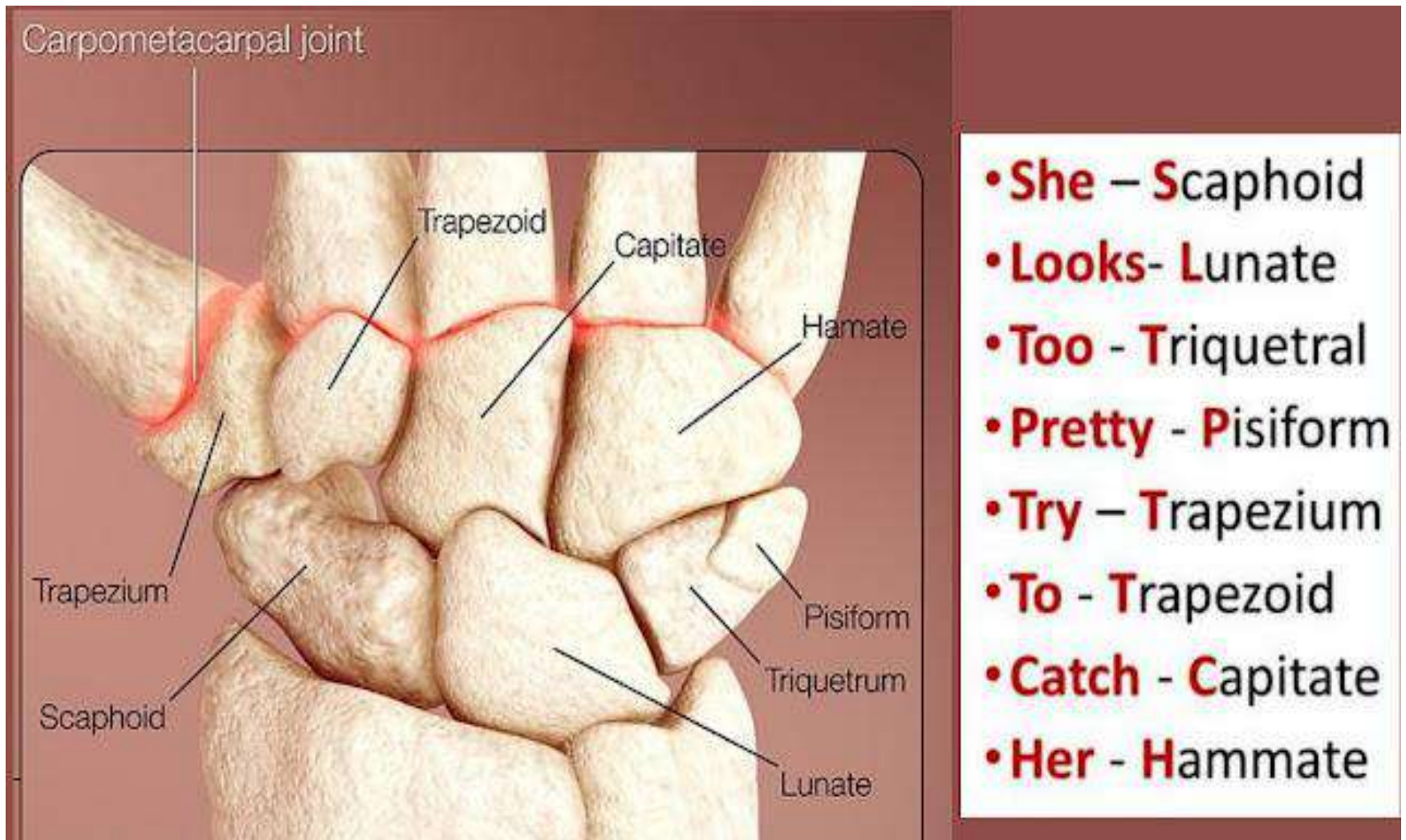
-The bases articulate with the distal row of carpal bones --
The heads articulate with the bases of proximal phalanges .
The shaft is slightly concave forward.

Bones of the fingers (14 Phalanges):

Each finger has three phalanges (proximal, middle, and distal), while the thumb has only two (proximal and distal).



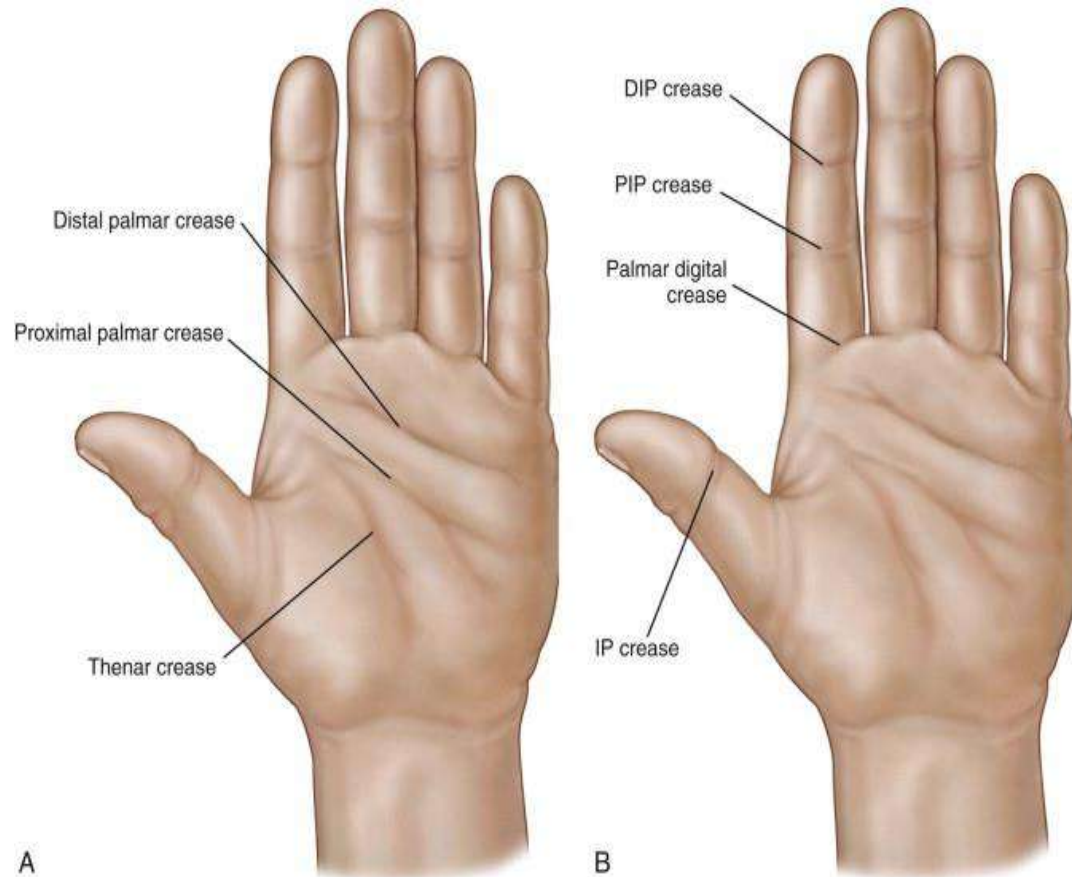
How to remember the carpal bones?



Skin of the palm of the hand

Thick, hairless, with creases and sweat glands.

-**Skin creases** : are thick folded skin over the joints .

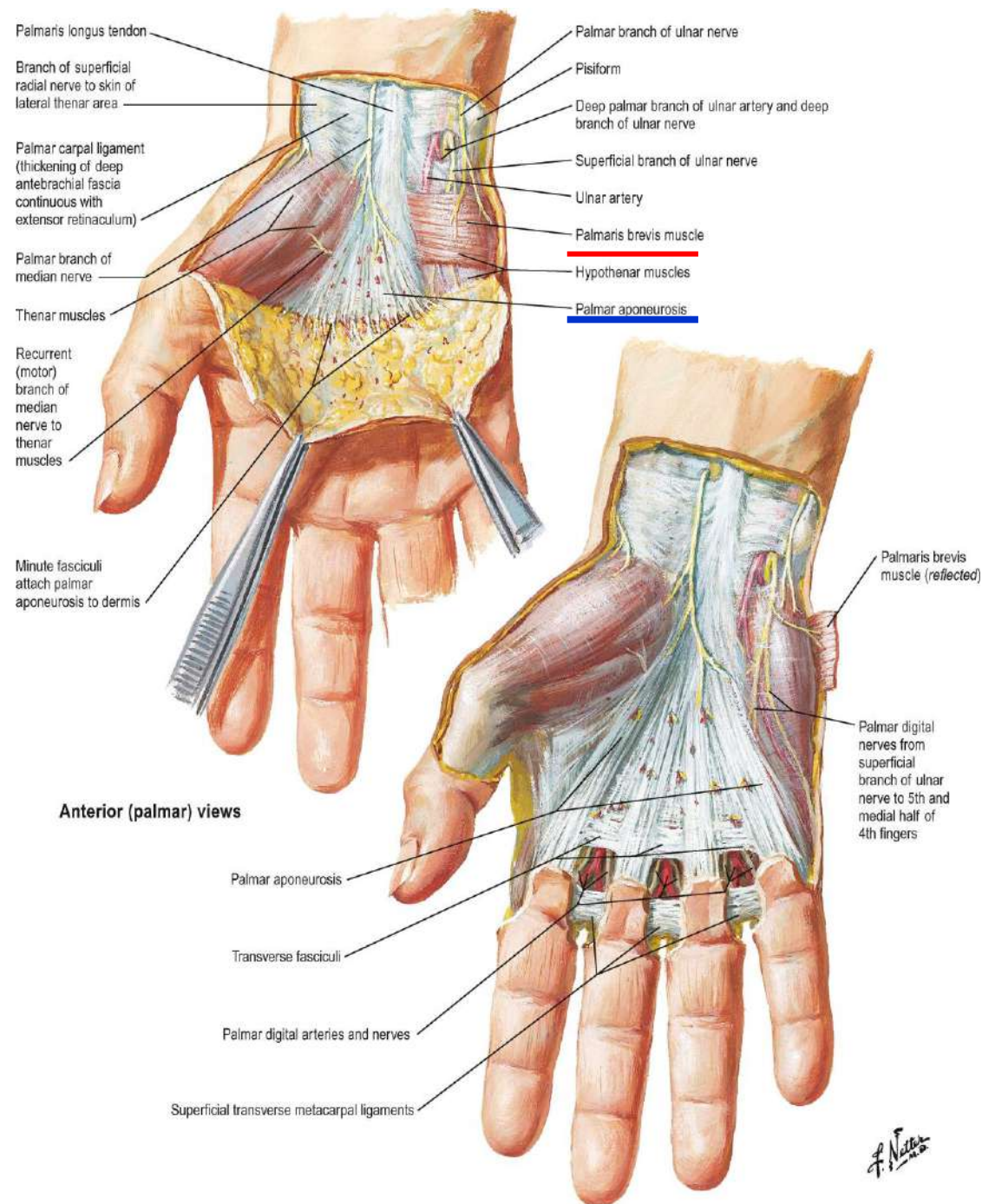


N.B. While the SKIN of the **dorsum of the hand** is thin, hairy, freely mobile on the underlying bones and tendons.

Fascia of the Hand

I. Superficial fascia contains the **Palmaris brevis** muscle

II. Deep fascia is called the **Palmar aponeurosis**.



Deep Fascia of the hand

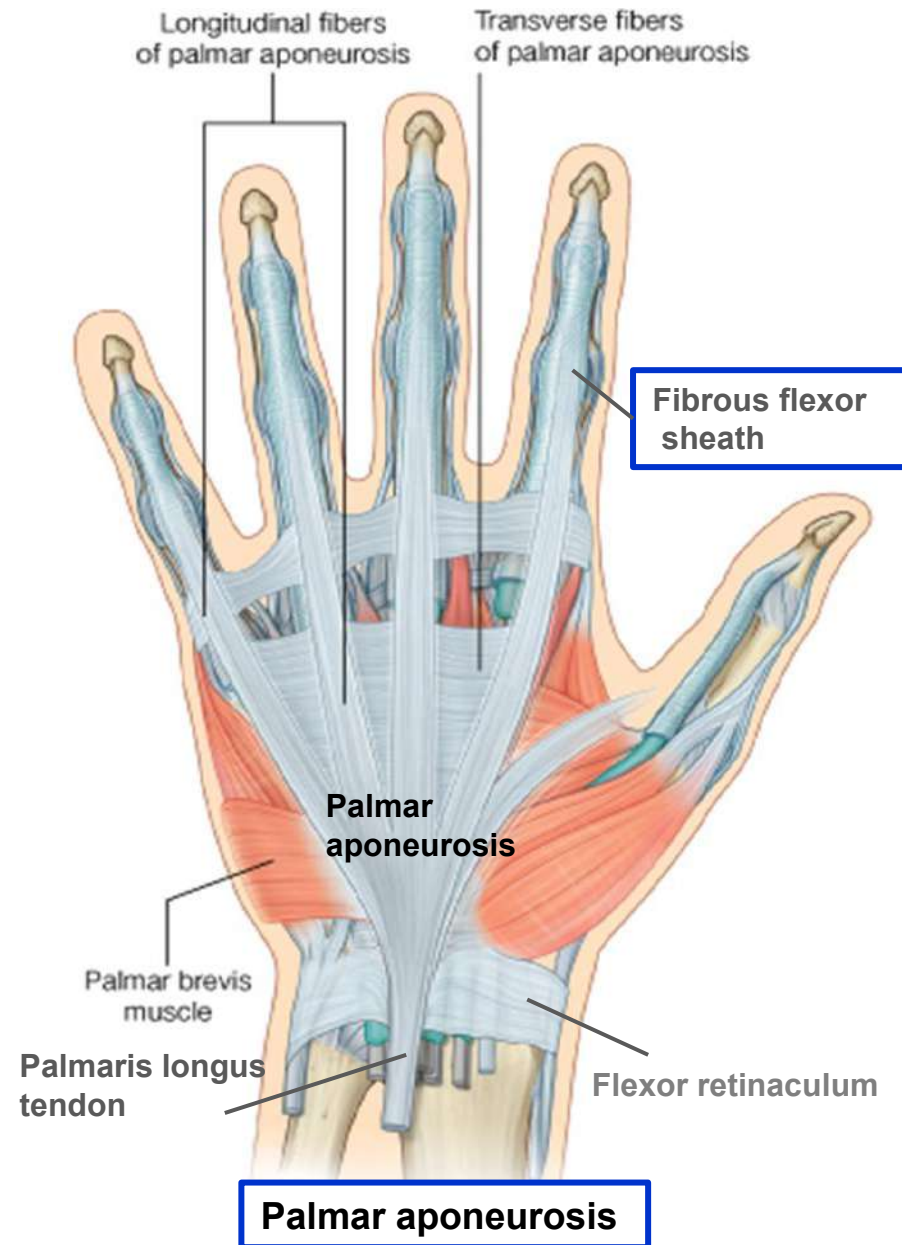
1. Palmar aponeurosis

It is a **triangular thickening of deep fascia** in the palm, and is continuous with the **palmaris longus tendon**, the **thenar and hypothenar fasciae** and the **flexor retinaculum**

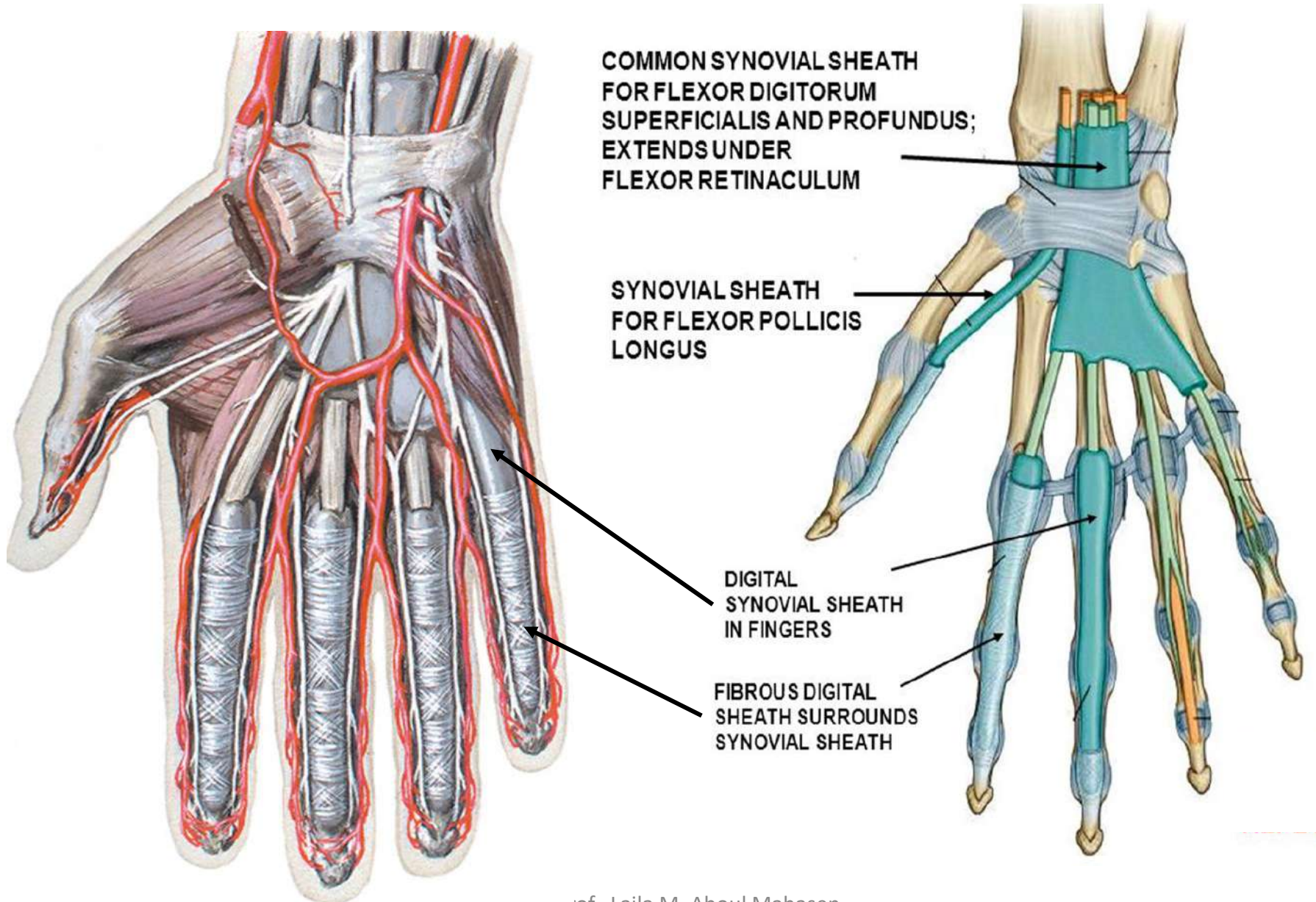
-At the base of medial 4 fingers ,it divides into 4 slips .Each slip divides into 2 bands which joins the **fibrous flexor sheaths** . It protects the **superficial palmar arterial arch** and the **palmar digital nerves**.

2-Fibrous flexor sheaths

They are **thickening of deep fascia in front of fingers** covering the **flexor tendons and their synovial sheaths**.



Fibrous flexor sheaths covering & protecting the synovial sheaths of the flexor tendons



Remember!

Flexors Retinaculum

Definition : it is a **thickening of deep fascia** that holds the long flexor tendons in position **at the wrist**.

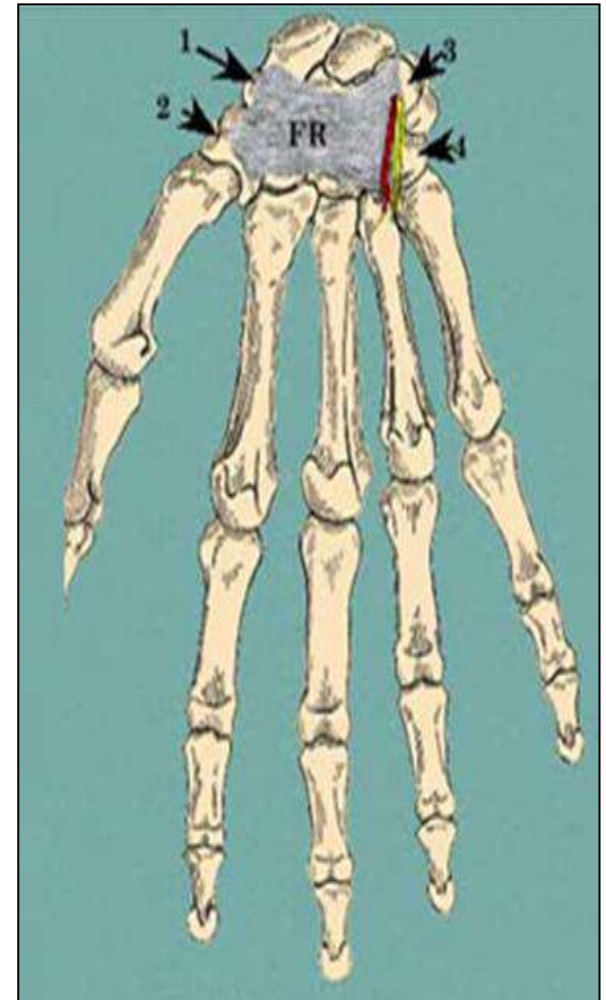
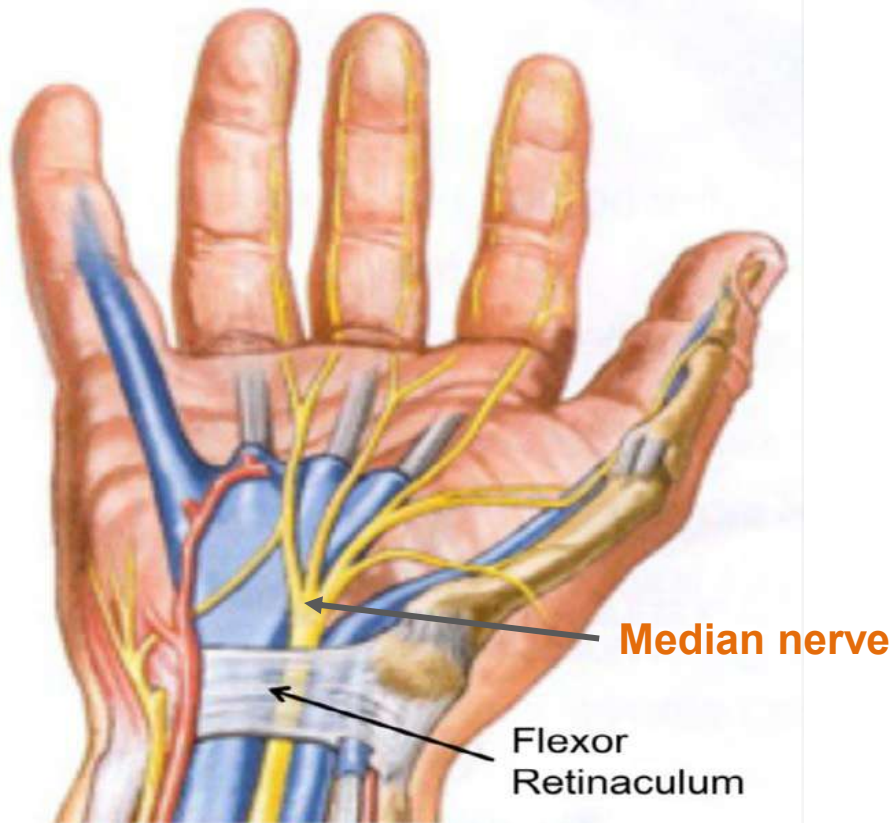
Importance :

The retinaculum converts the concave anterior surface of the carpal bones into the **carpal tunnel** for the passage of **Median nerve** & **flexors tendons** of fingers & thumb

Attachment:

Medially: to pisiform(3) , hamate(4)

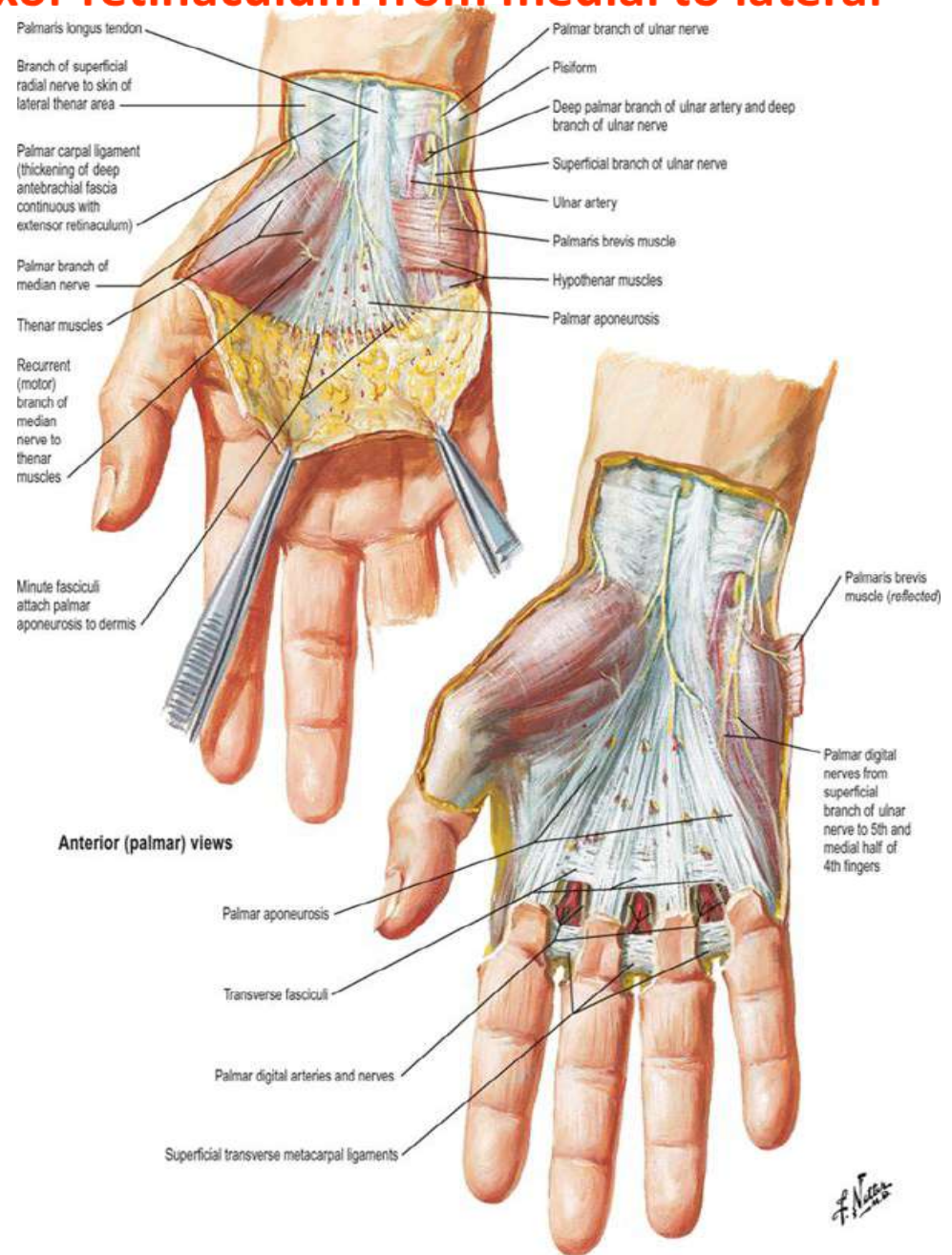
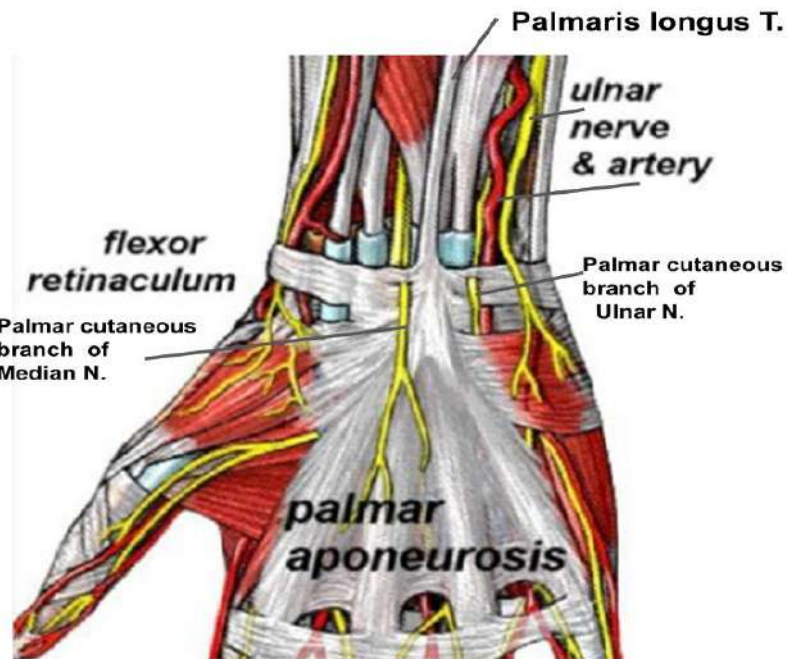
Laterally : scaphoid(1), trapezium(2)



Structures passing Superficial to Flexor retinaculum from medial to lateral

Remember!

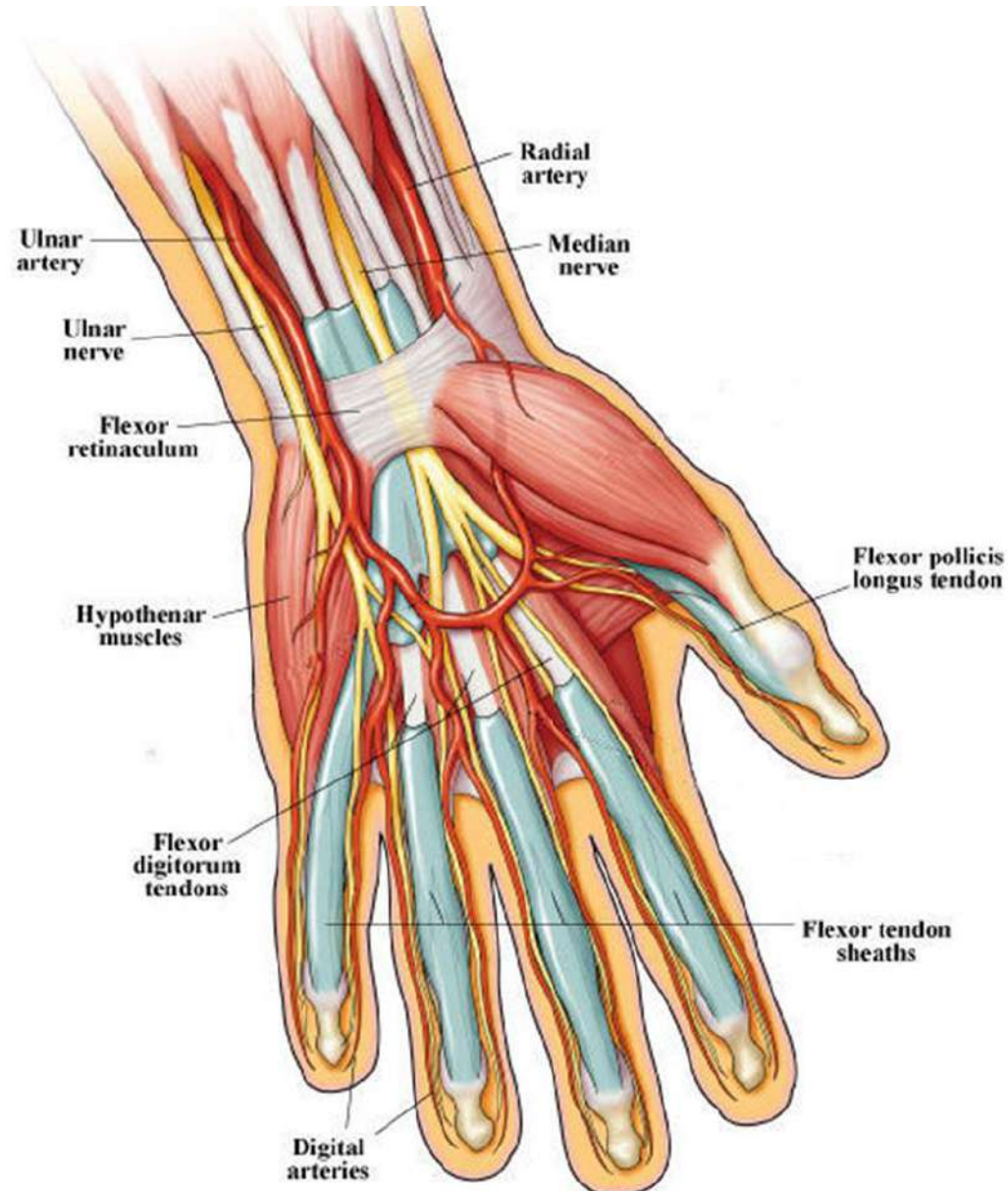
1. Tendon of flexor carpi ulnaris
2. Ulnar N.
3. Ulnar A.
4. Palmar cutaneous branch of Ulnar N.
5. Tendon of palmaris longus
6. Palmar cutaneous branch of Median N.



Remember!

Structures pass Deep to Flexor Retinaculum from medial to lateral :

- 1 -Tendons of flexor digitorum Superficialis & profundus within a common synovial sheath**
- 2- Median nerve**
- 3- Anterior Interosseous branch of median N.**
- 4- Ant. Interosseous artery**
- 5-Tendon of flexor pollicis longus & its synovial sheath**
- 6-Tendon of flexor carpi radialis & its synovial sheath**

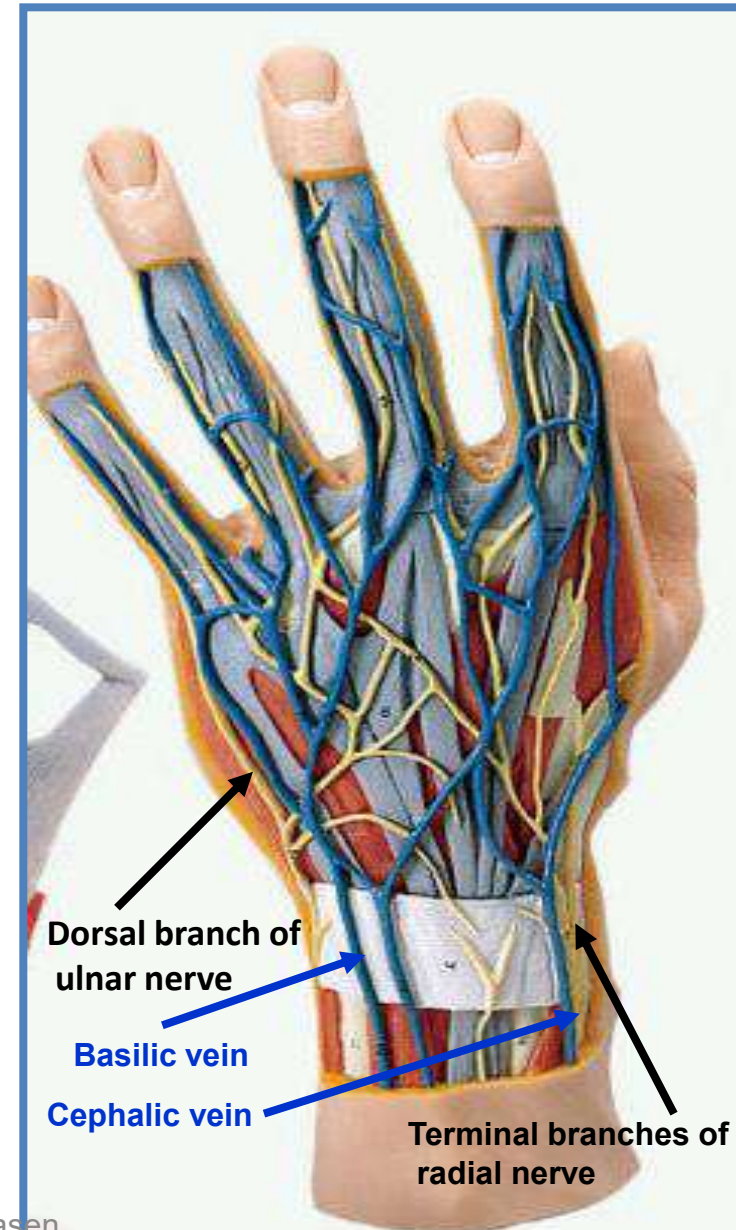
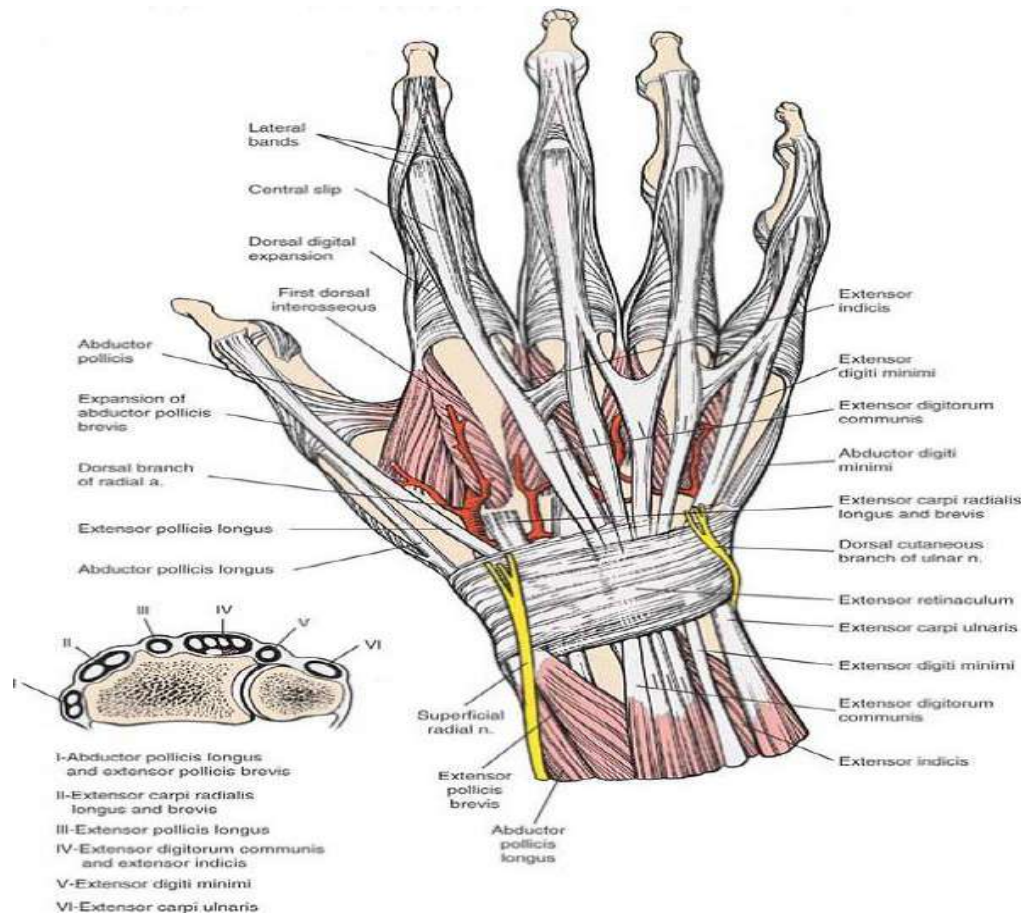


Remember!

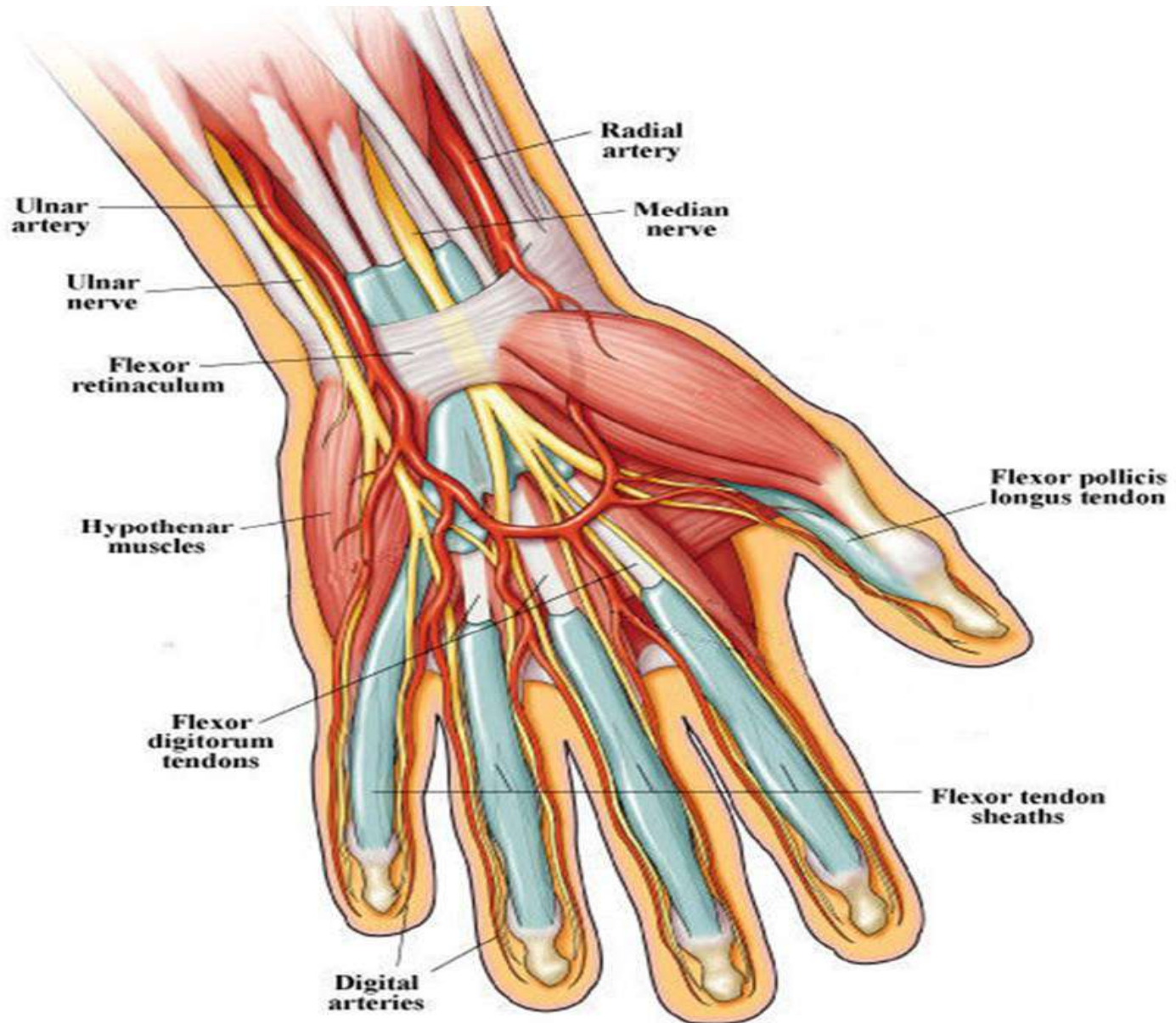
Extensor Retinaculum

Structures **superficial** to extensor retinaculum of the hand :

- 1- Beginning of cephalic & basilic veins
- 2- Superficial branches of radial nerve
- 3- Dorsal branch of ulnar nerve



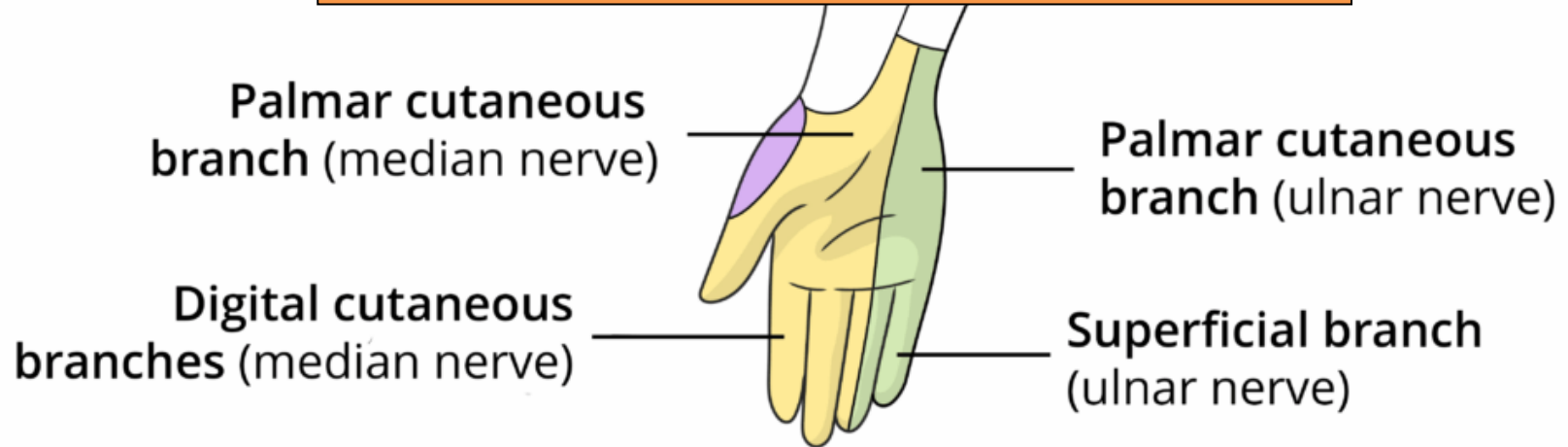
Nerves & vessels of the hand



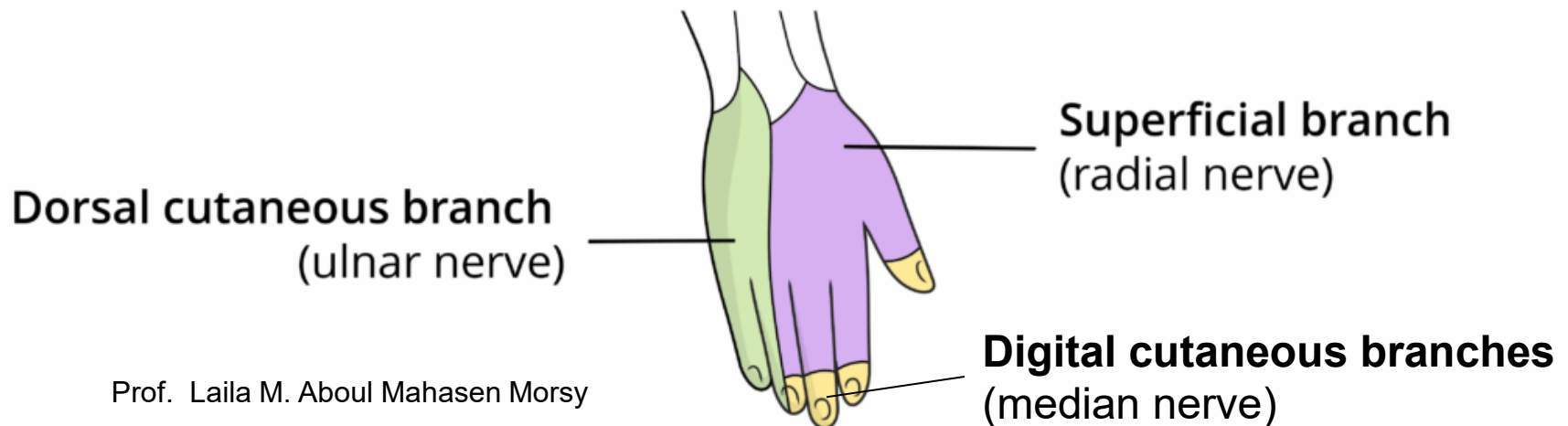
Cutaneous supply of the hand



Cutaneous nerve supply of Palm of the hand



Cutaneous nerve supply of Dorsum of the hand



Cutaneous supply of hand is done by branches of 3 nerves :

Median N. - Ulnar N. - Radial N.

1. Median nerve supplies skin over:

I. Palmar cutaneous branch. : Lateral 2/3rd of palm

II. Palmar digital branches : Palmar surface, nails and dorsal surface of distal phalanx of lateral 3 1/2 digits including thumb)

2. Ulnar nerve supplies skin over :

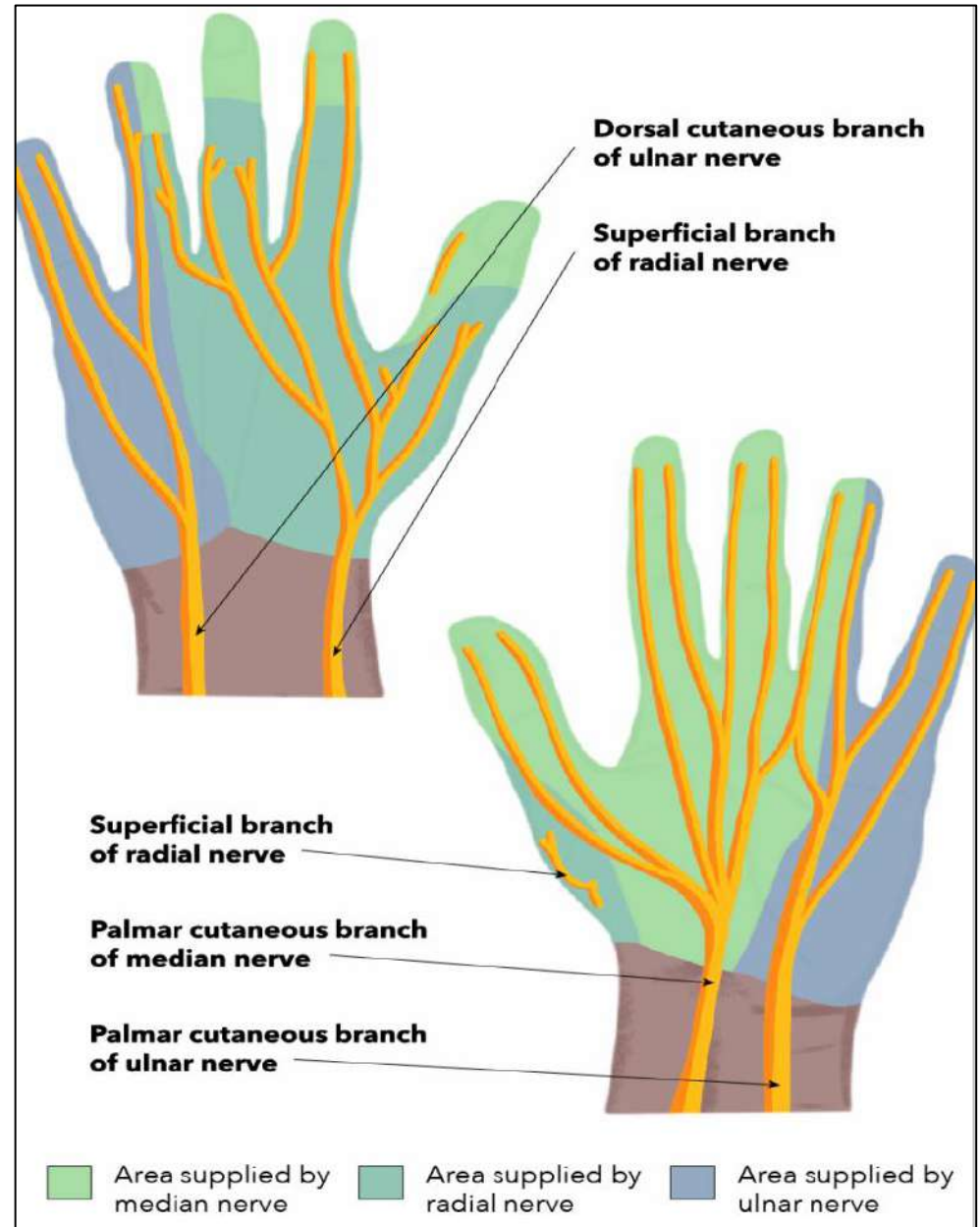
I. Palmar cutaneous branch : Medial 1/3rd of palm

II. Dorsal cutaneous branch : Medial 1/3rd of dorsum of hand (arises in forearm)

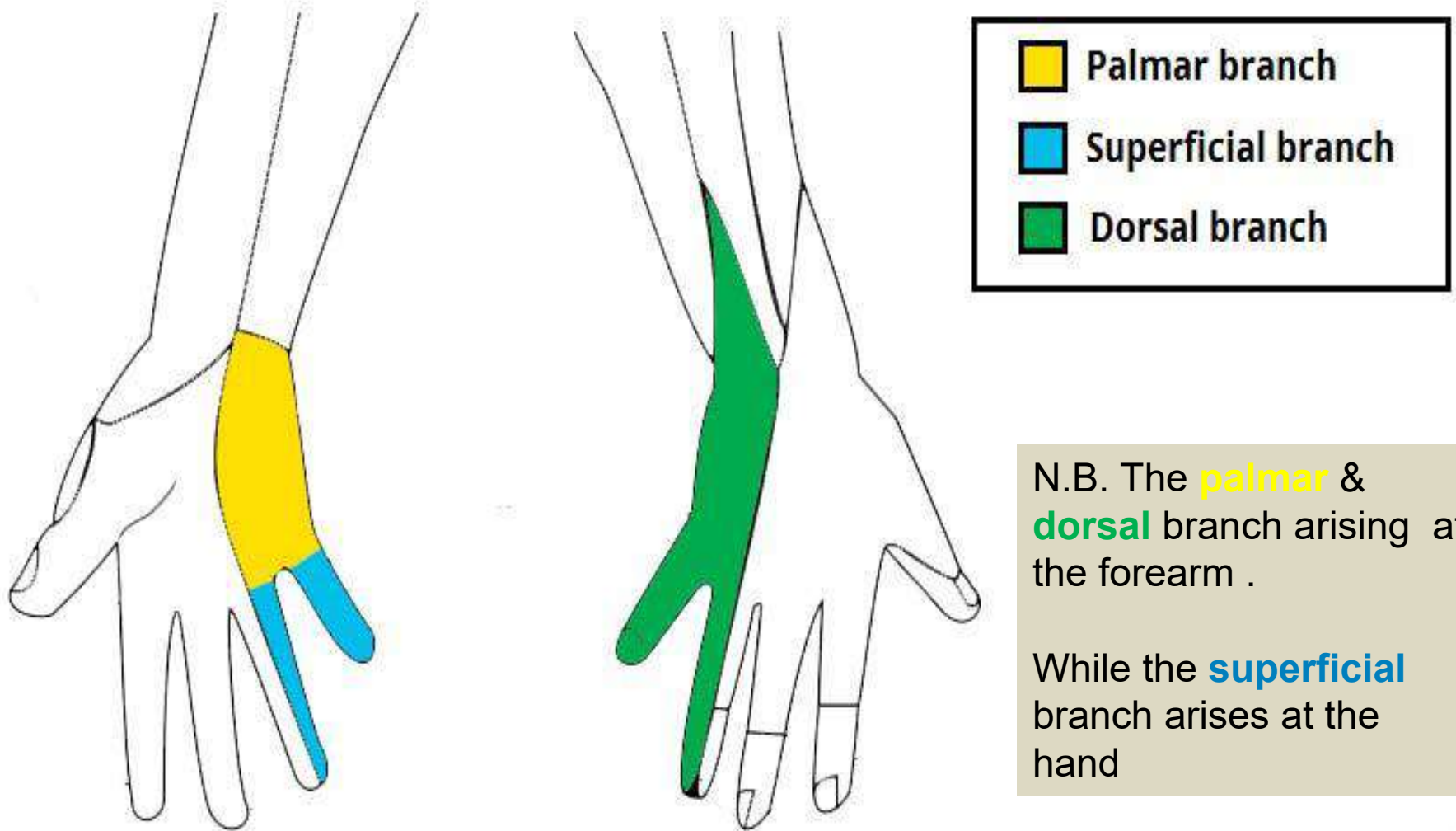
III. Superficial and dorsal digital branches: Palmar and dorsal surfaces of medial 1 1/2 digits

3. Radial nerve supplies skin over:

Superficial branch : Lateral 2/3rd of dorsum of hand including dorsal surface of proximal and middle phalanges of lateral 3 1/2 digits.



Cutaneous supply of ulnar nerve



Ulnar nerve in the hand (mainly muscular)

In the forearm : it gives :

Palmar cutaneous branch : to the skin over the hypothenar region.

Dorsal cutaneous branch : to supply the skin of medial 1/3 of dorsum of the hand

Terminal branches in the hand:

A. Superficial terminal branch : (motor & sensory)

1. **Motor** : **palmaris brevis**

2- **Sensory** : **Superficial terminal branches** : give **3 palmar digital branches** : to 5th and medial 1/2 of 4th finger.

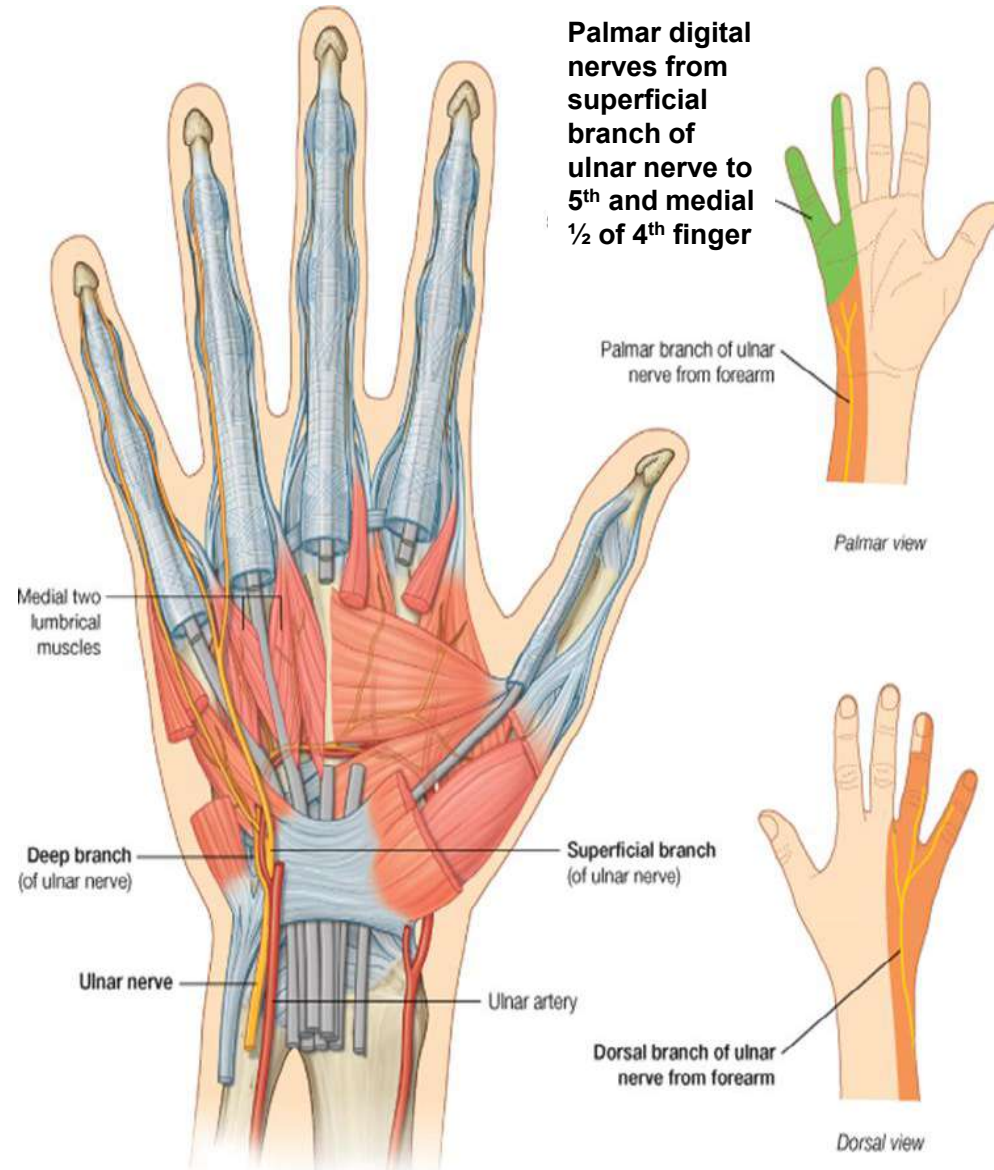
B . Deep terminal branch: (motor) :

1. **Hypothenar muscles**

2. **Medial two lumbricals**

3. **All interossei**

4. **Adductor pollicis** .



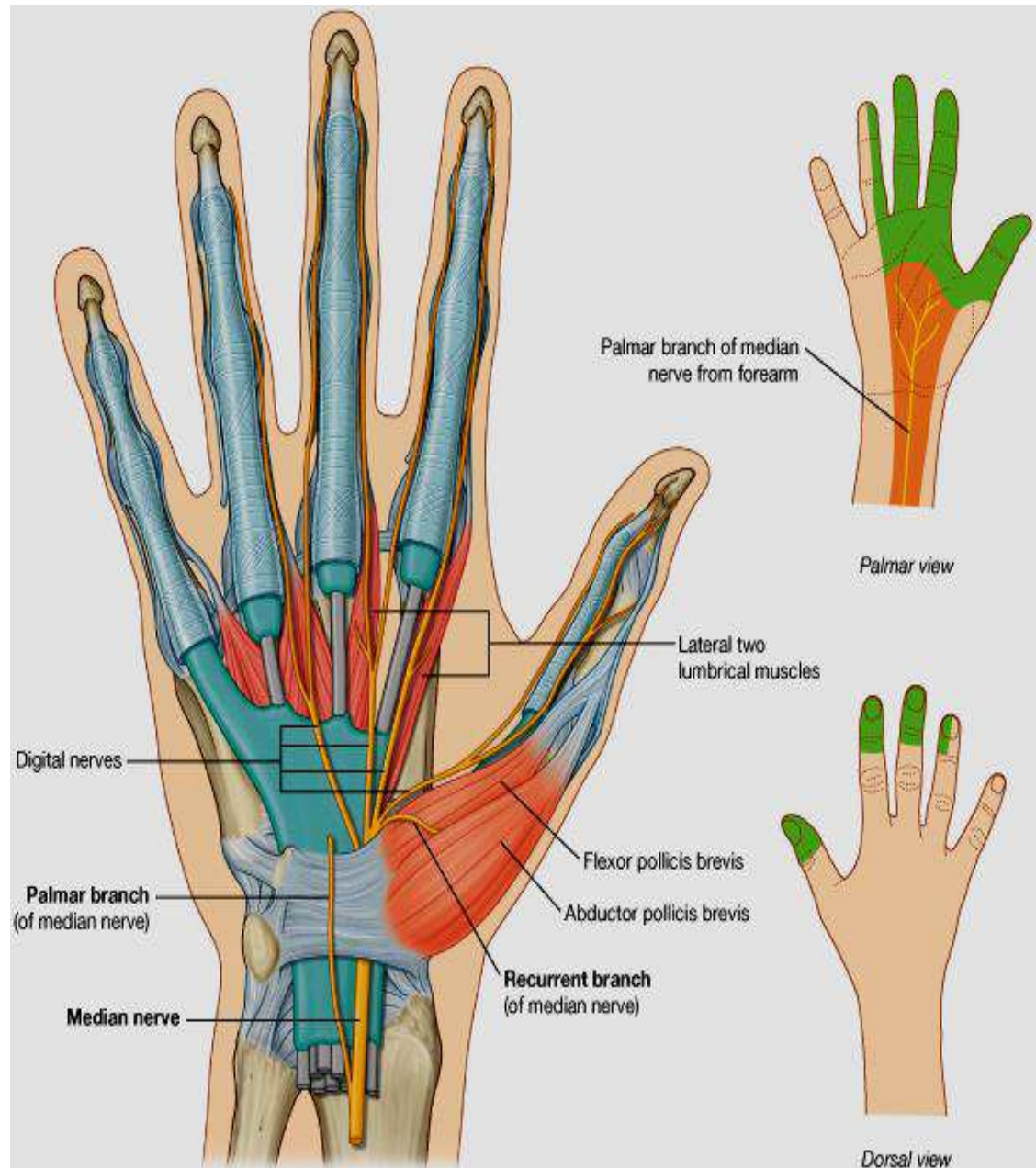
Median nerve in the hand (mainly cutaneous)

- **In the forearm** : it gives the **Palmar cutaneous branch** : it passes **superficial** to **flexor retinaculum** to supply the skin of the **lateral 2½ of the palm**

In the hand :

1- Muscular : **3 thenar muscles & lateral two lumbricals** .

2- Cutaneous : **5 palmar digital branches** to the **palmar surface** of the **lateral 3½ fingers**), as well as the **dorsal surface** of these fingers to a variable degree (**to distal & middle phalanges**)



Superficial branch of radial nerve

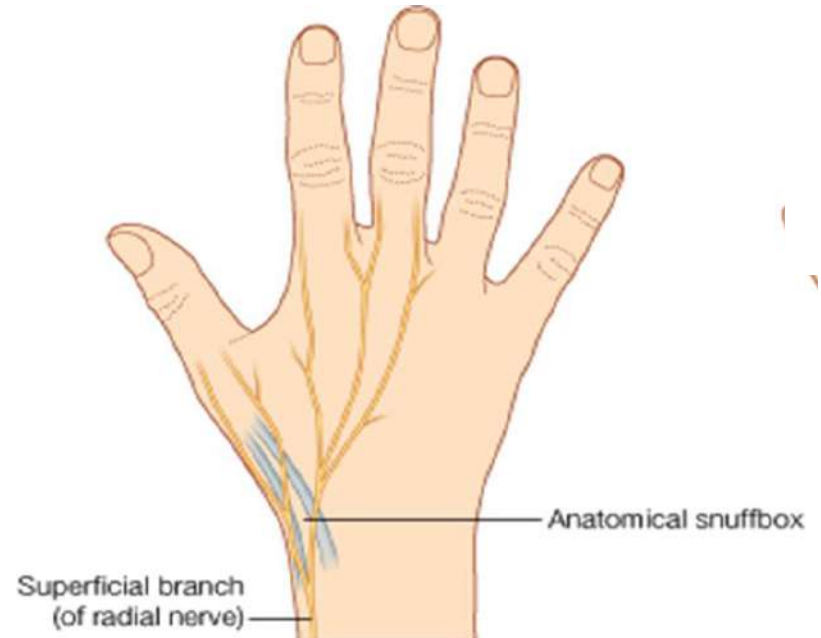
Superficial branch (cutaneous):

is considered as a direct continuation of radial nerve

Course :It descends in the forearm under cover of the brachioradialis muscle in contact with radial artery .It runs distally to the dorsum of the hand

Branches :

1. Skin of the root of the thumb
2. lateral $\frac{2}{3}$ of the back of the hand and the lateral $3\frac{1}{2}$ fingers (except the skin of back of distal phalanges which supplied by the median nerve) .



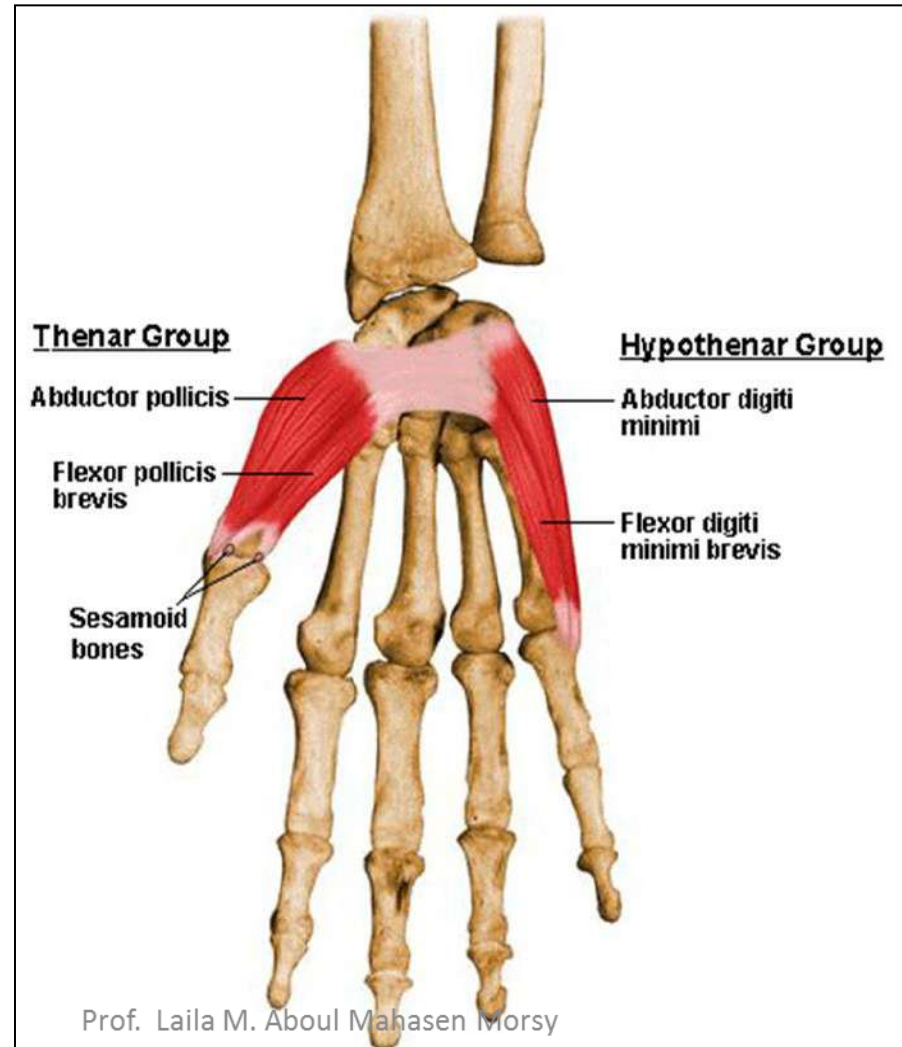
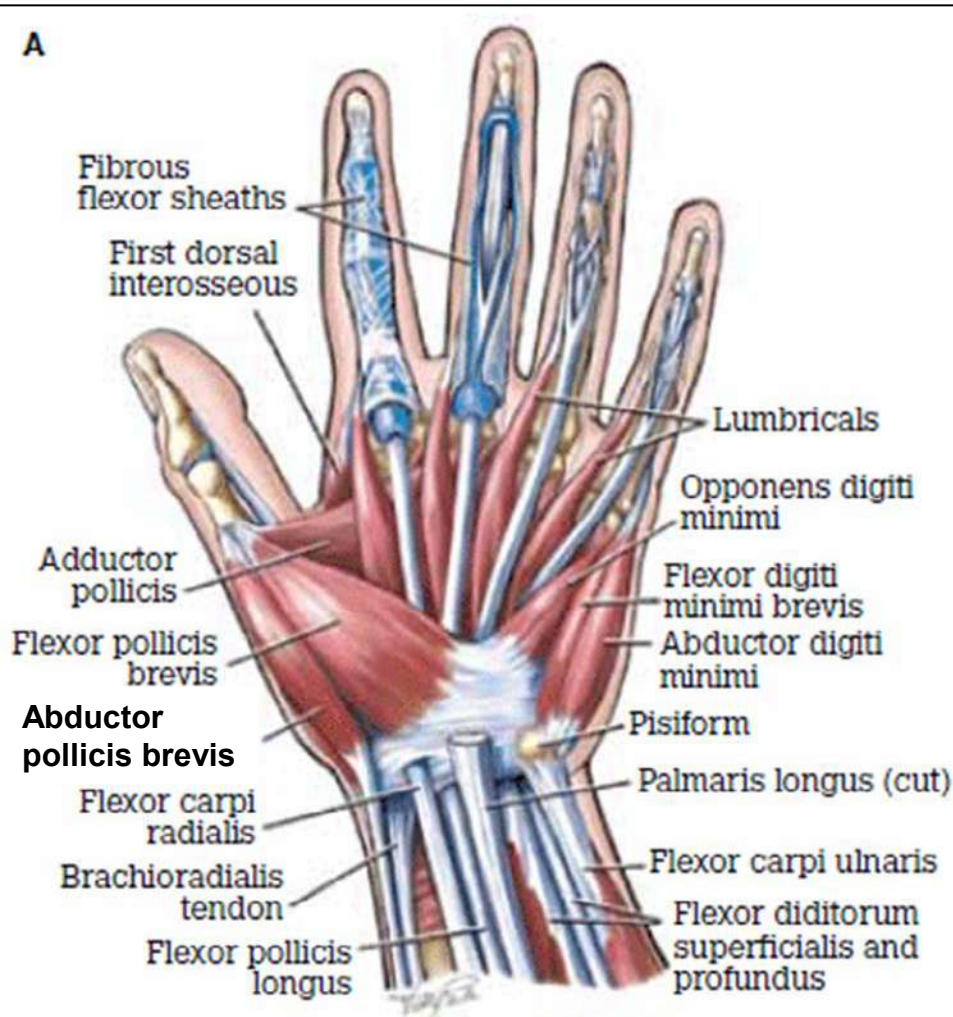
Palmar view



Dorsal view

Intrinsic Muscles of the Hand (3 Main compartments)

1. Muscles of **thenar** eminence(4 muscles)
2. Muscles of **hypothenar** eminence (3 muscles) –Palmaris brevis
3. Muscles of **intermediate** region (lumbricals & interossei)



Intrinsic muscles of the hand

Palmaris brevis muscle

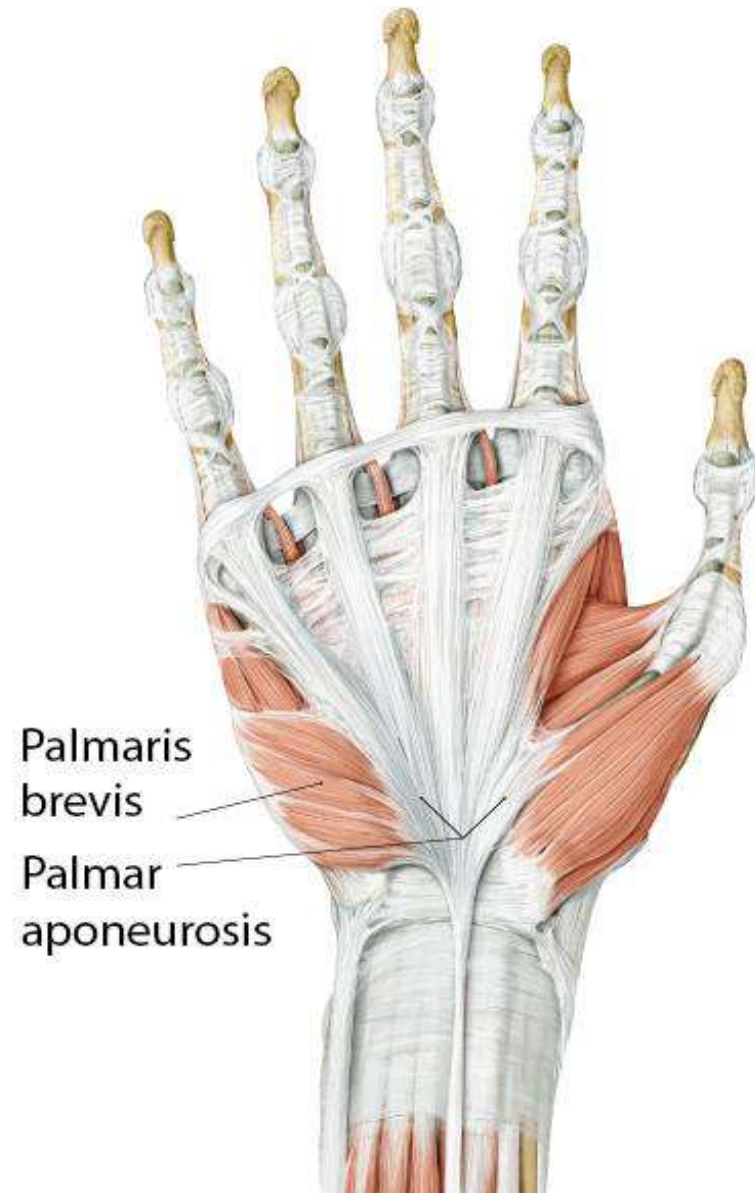
It is a small, hand muscle lies on superficial fascia , on ulnar side of the palm, **superficial** to the **hypothenar muscles**.

It originates from the palmar aponeurosis and flexor retinaculum and inserts into the skin of the medial border of the hand.

Action :

1. wrinkle the skin of the hypothenar eminence to deepen the palm for a more effective grip
2. Protect the ulnar nerve and artery from compression

Nerve supply : superficial branch of the ulnar nerve,



Muscles of thenar eminence

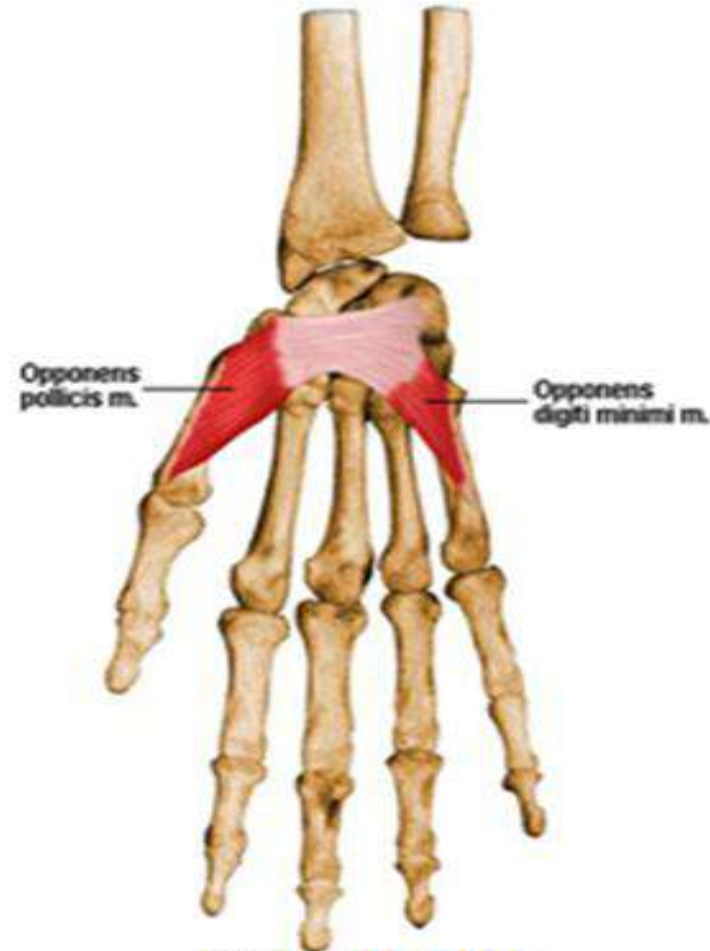
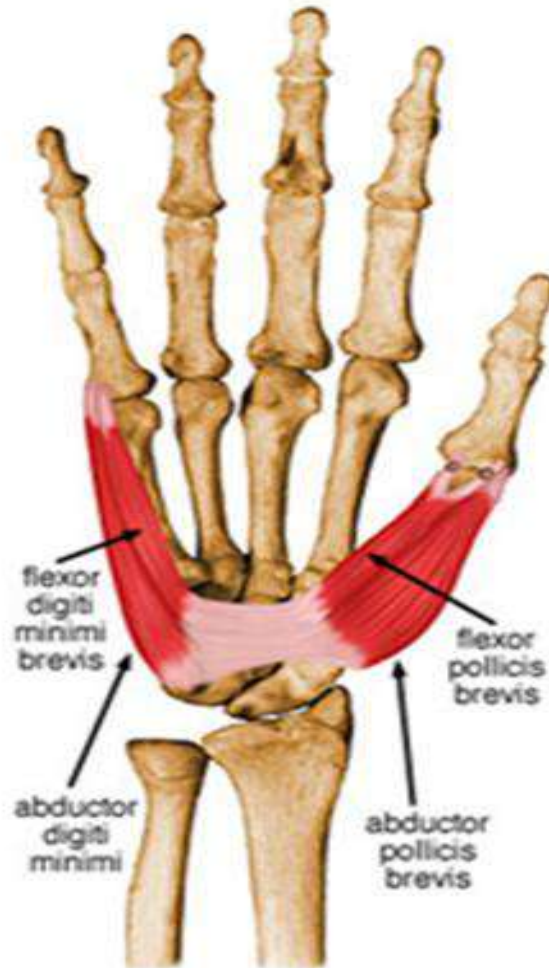
1. Abductor pollicis brevis
2. Flexor pollicis brevis
3. Opponens pollicis(deep)
4. Adductor pollicis



Adductor Pollicis

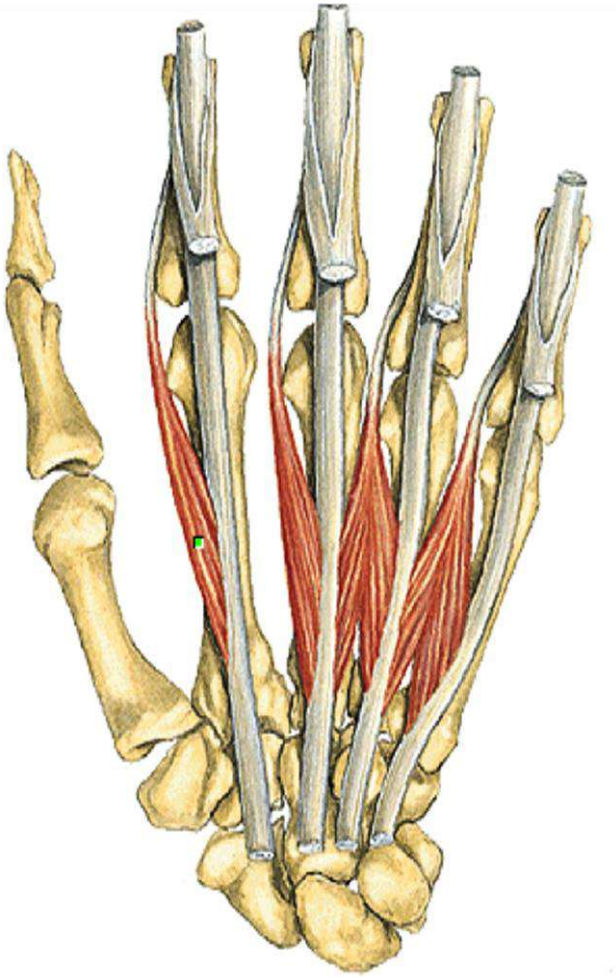
Muscles of hypothenar eminence

1. Abductor digiti minimi
2. Flexor digiti minimi
3. Opponens digiti minimi(deep)



Deep muscles

Muscles of intermediate region (lumbricals & interossei)

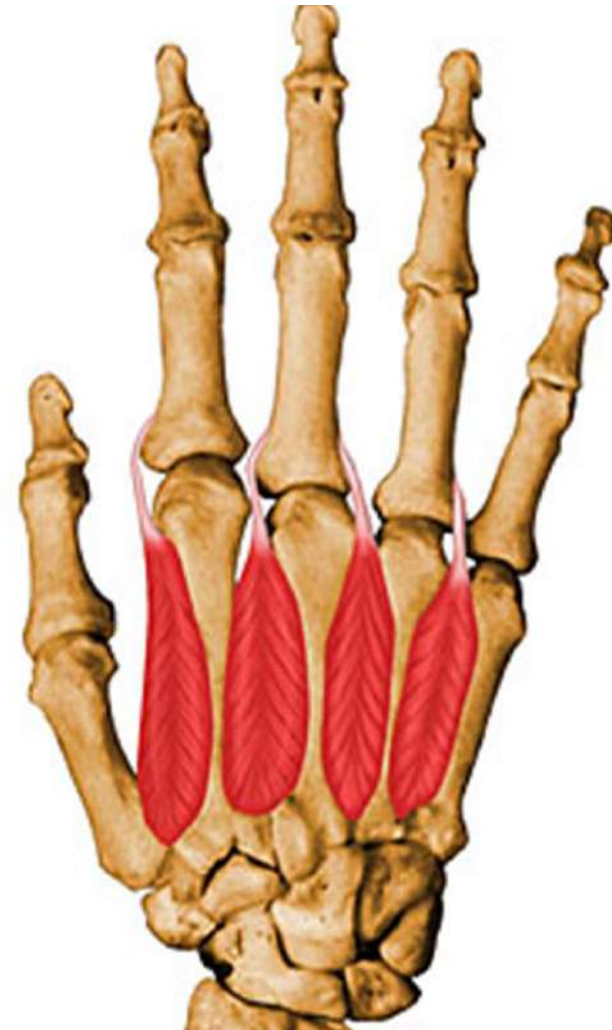


4 Lumbricals
(writing position)



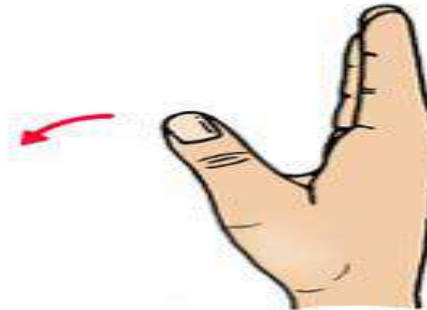
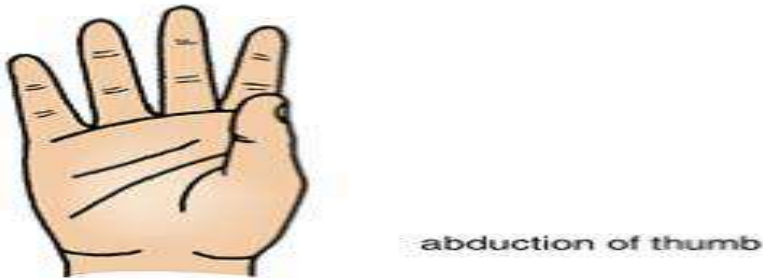
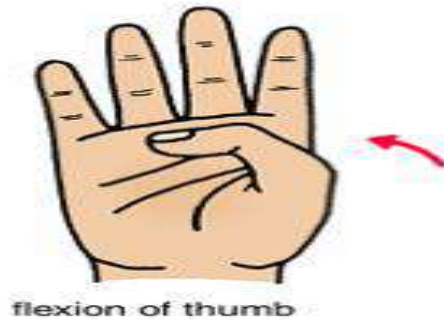
3 Palmar interossei
(Adductors)

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4 Dorsal interossei
(Abductors)

Movements of the thumb

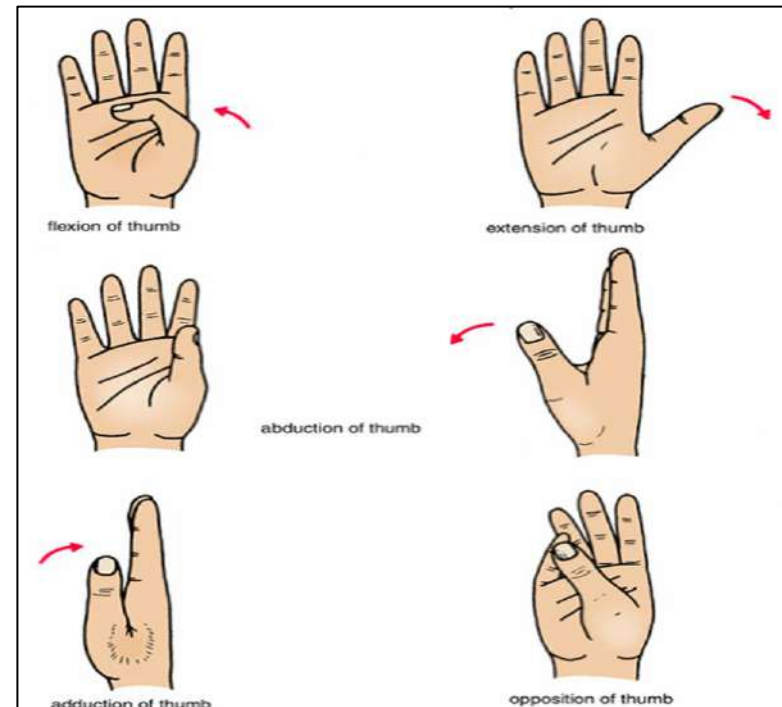
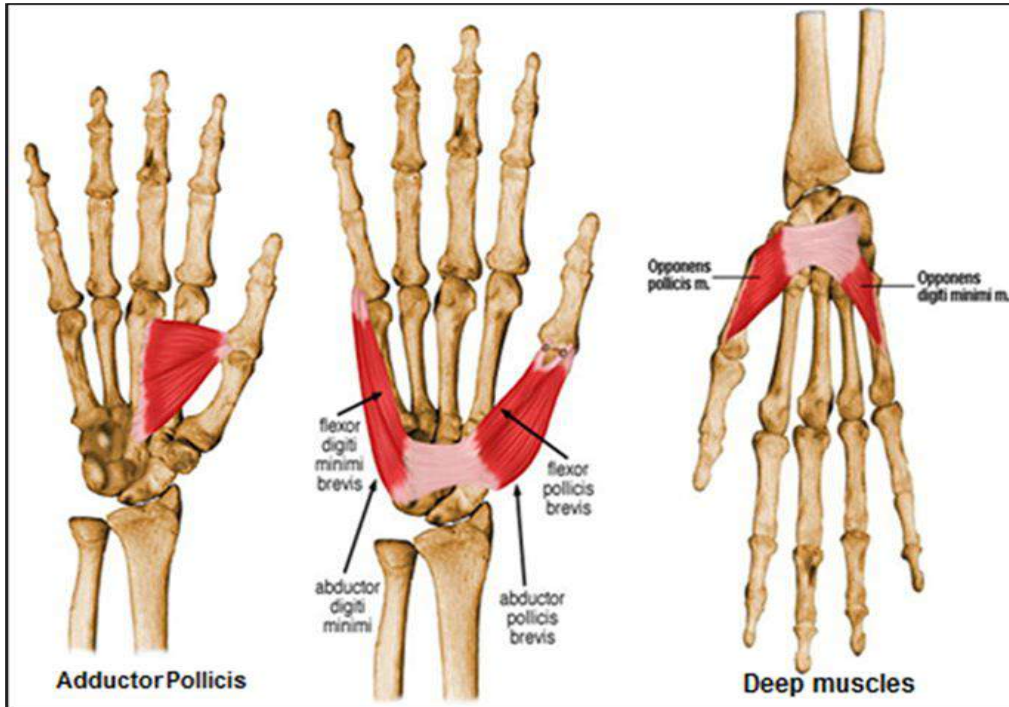


Opposition of thumb : pulls thumb medially & forward across palm

Opposition of little finger : pulls the little finger toward the thumb laterally & across palm to cup the palm

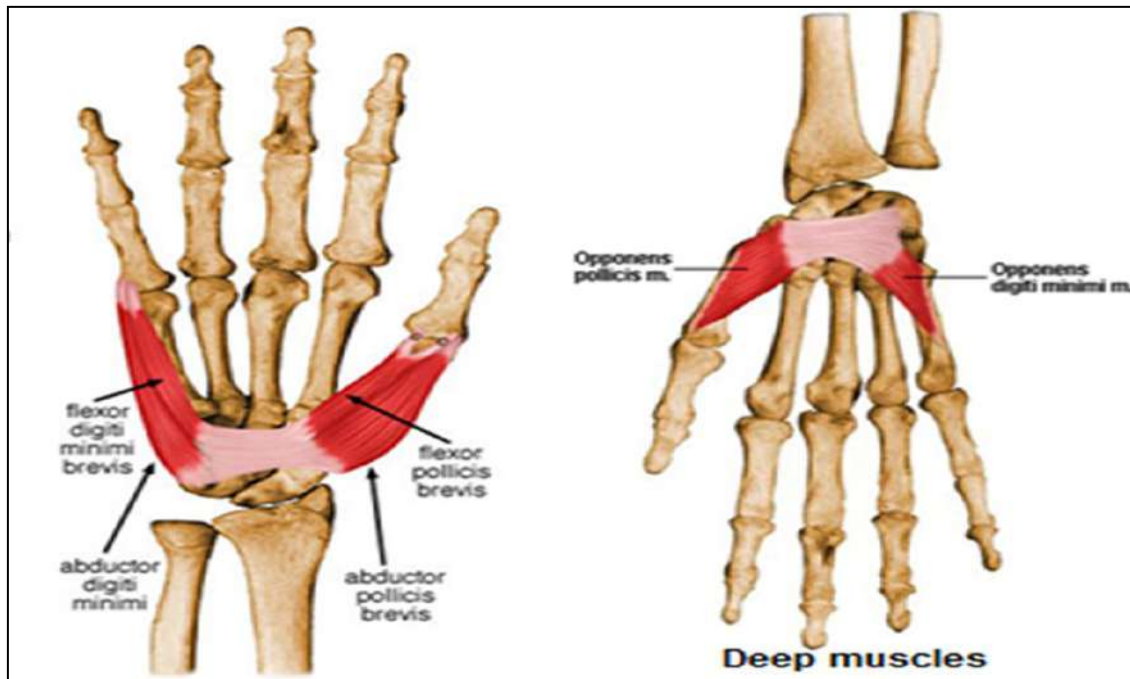
Muscles of thenar eminence (4 Ms.)

Muscle	Nerve Supply	Action
Abductor pollicis brevis	Median nerve	Abducts the thumb
Flexor pollicis brevis	Median nerve	Flexes the thumb at 1 st metacarpophalangeal joint(MPJ)
Opponens pollicis (deep)	Median nerve Prof. Laila M. Aboul Mahasen	Opposes the thumb toward the other fingers medially to cup the palm (strong grip of objects)
Adductor pollicis	Deep branch of Ulnar nerve	Adducts the thumb



Muscles of hypothenar eminence

Muscle	Nerve Supply	Action
Abductor digiti minimi	Deep branch of Ulnar nerve	Abducts the little finger
Flexor digiti minimi	Deep branch of Ulnar nerve	Flexes the little finger at 5 th metacarpophalangeal joint(MPJ)
Opponens digiti minimi(deep)	Deep branch of Ulnar nerve	Opposes the little finger toward the thumb laterally by flexing and laterally rotated the 5 th MCP to cup the palm (strong grip of objects)

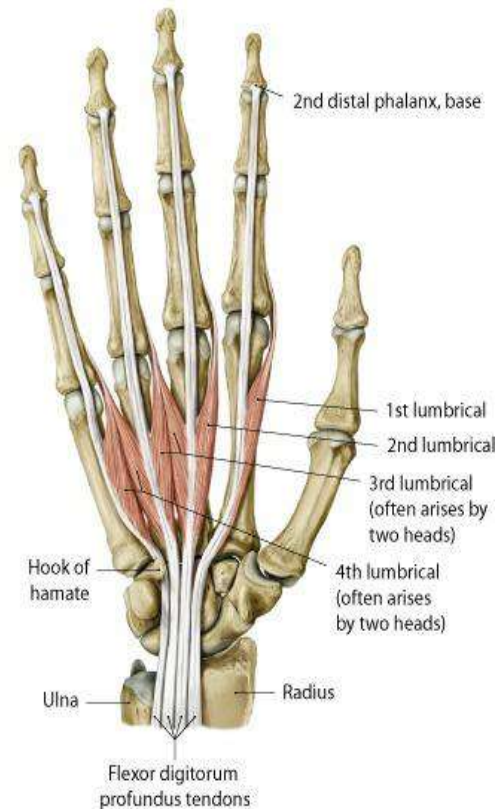
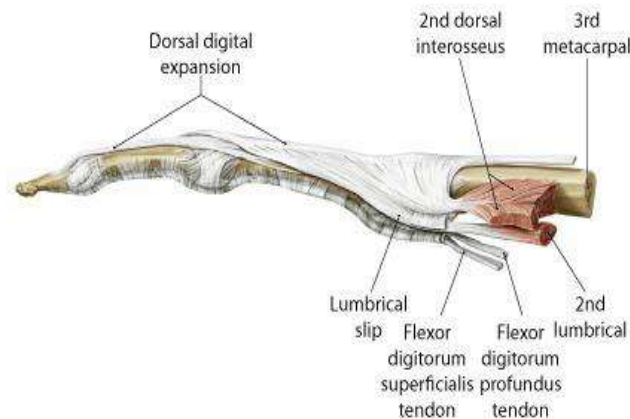
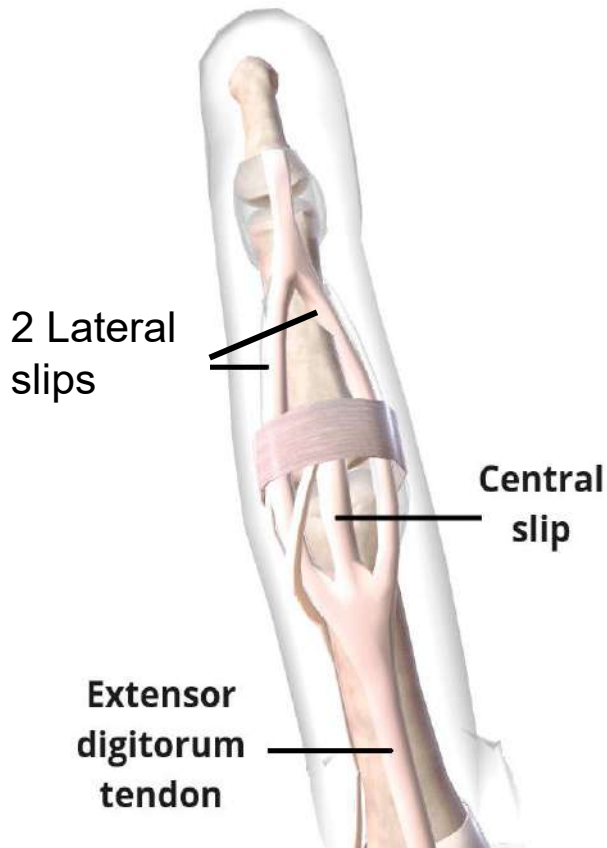


Extensor expansions

It is the insertion pattern of the 4 tendons of extensor digitorum.

Each tendon flatten dorsal to its corresponding proximal phalanx , then the flattened tendon is divided into 3 slips (band):

1. Central slip inserted into the base of the middle phalanx
2. 2 lateral slips that unite to form the terminal slip to the distal
3. The proximal part of the expansion receives the **Lumbricals & Interossei muscles**



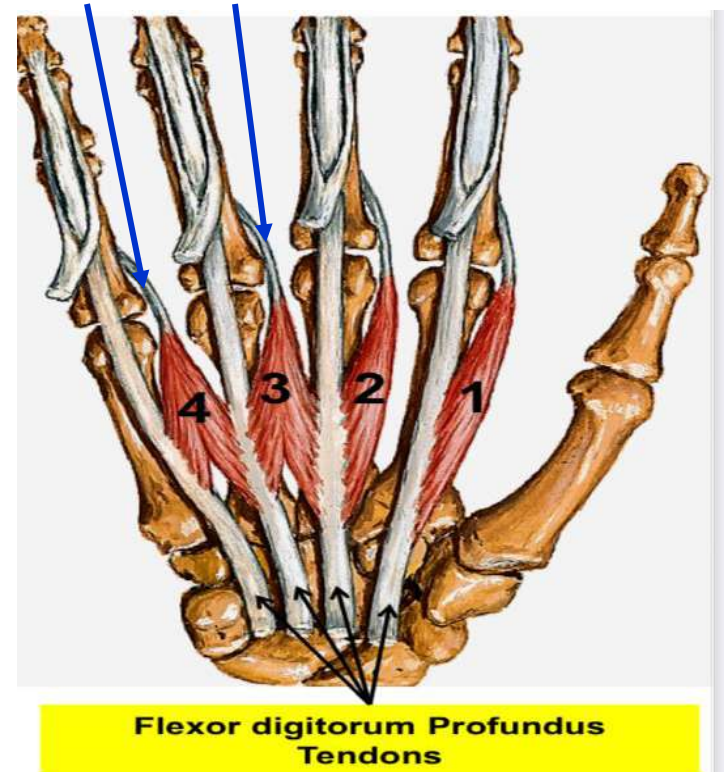
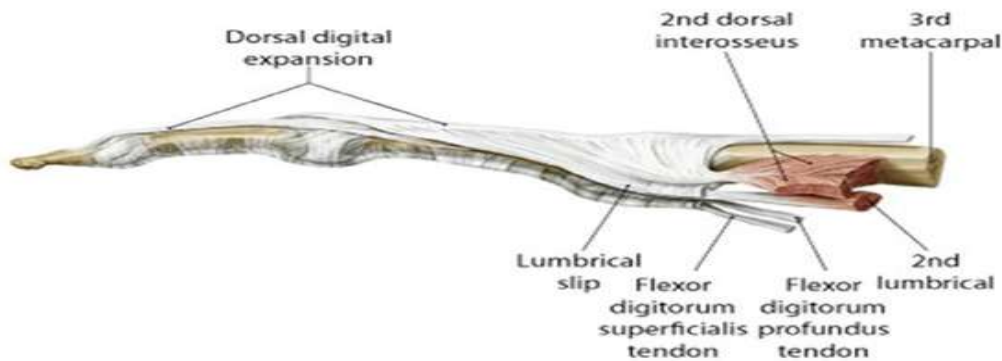
Muscles of intermediate region (lumbricals Muscles)

4 Lumbricals

- | | |
|------------------------------|--|
| 1-Median
(two
lateral) | Flex
metacarpophalangeal
joints and extend
interphalangeal joints |
| 2-ulnar
(two
medial) | |



Lumbricals & interossei muscles inserted in the proximal part of the extensor expansion



Muscles of intermediate region (interossei)

4 Dorsal interossei

Ulnar

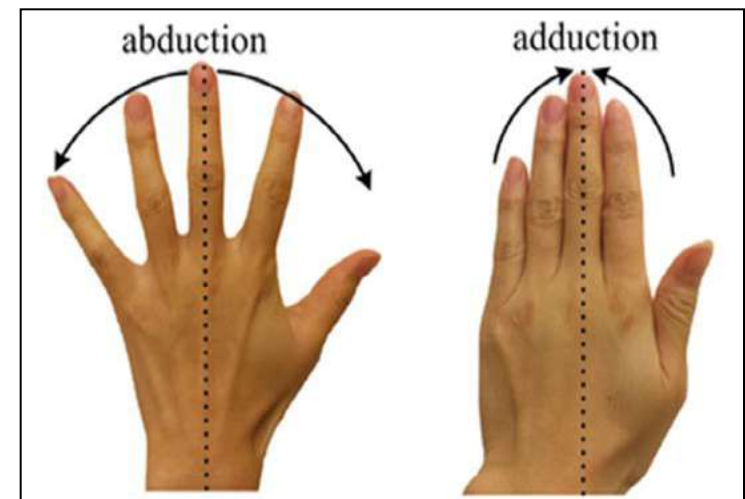
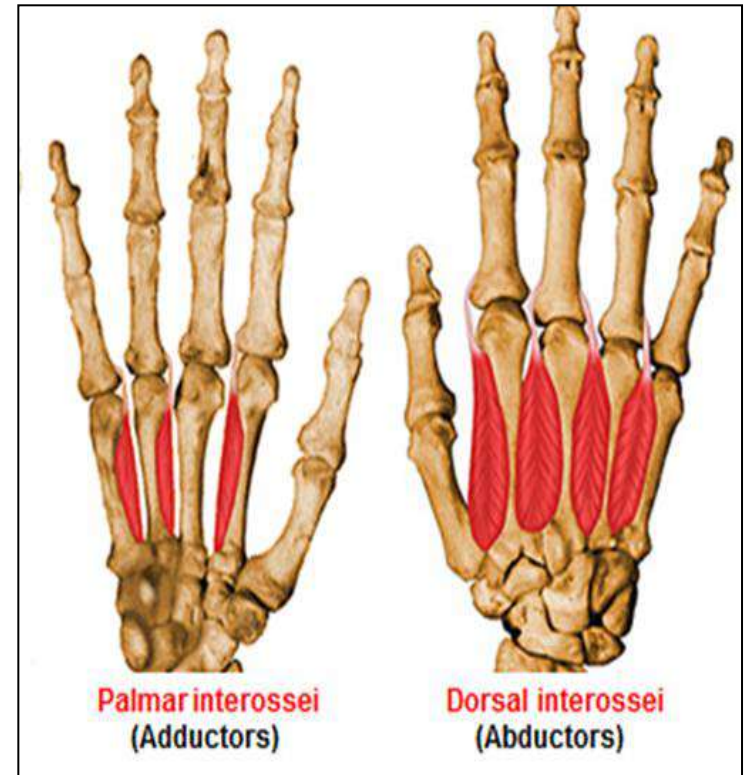
- 1-Abduct fingers .
- 2- Put the hand in writing position (by flexing metacarpophalangeal joints & extending interphalangeal joints)

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4 or 3 Palmar interossei

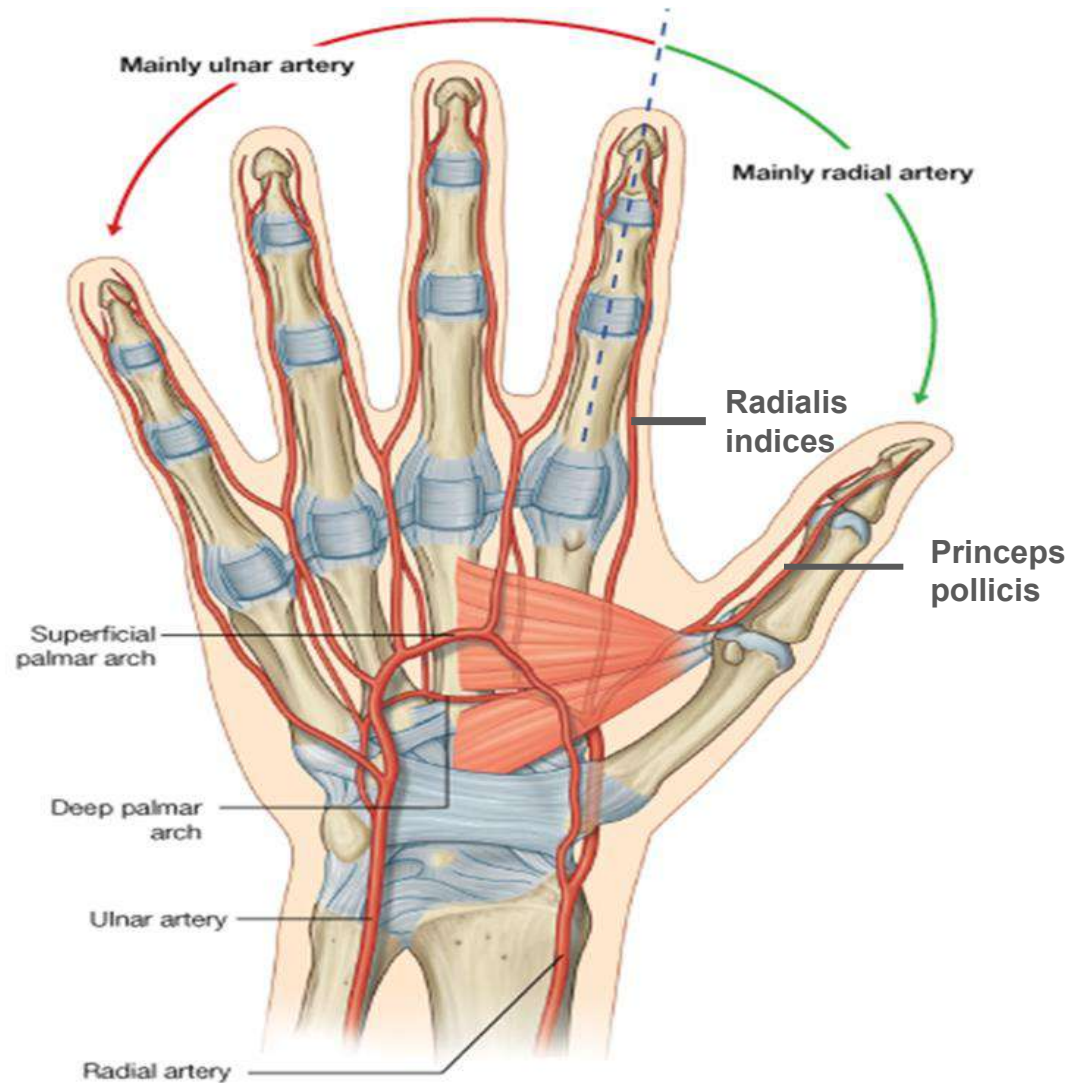
Ulnar

- 1-Adduct fingers.
- 2-Flex metacarpophalangeal joints; extend interphalangeal joints

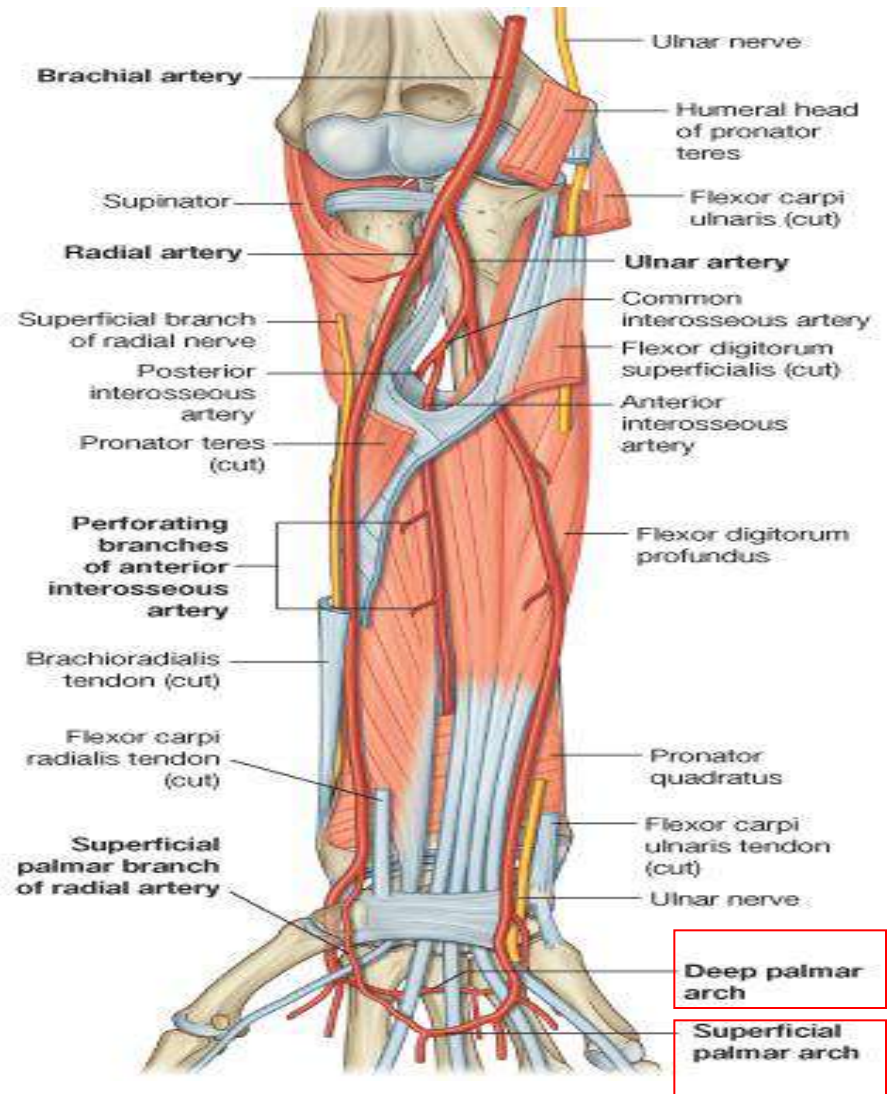
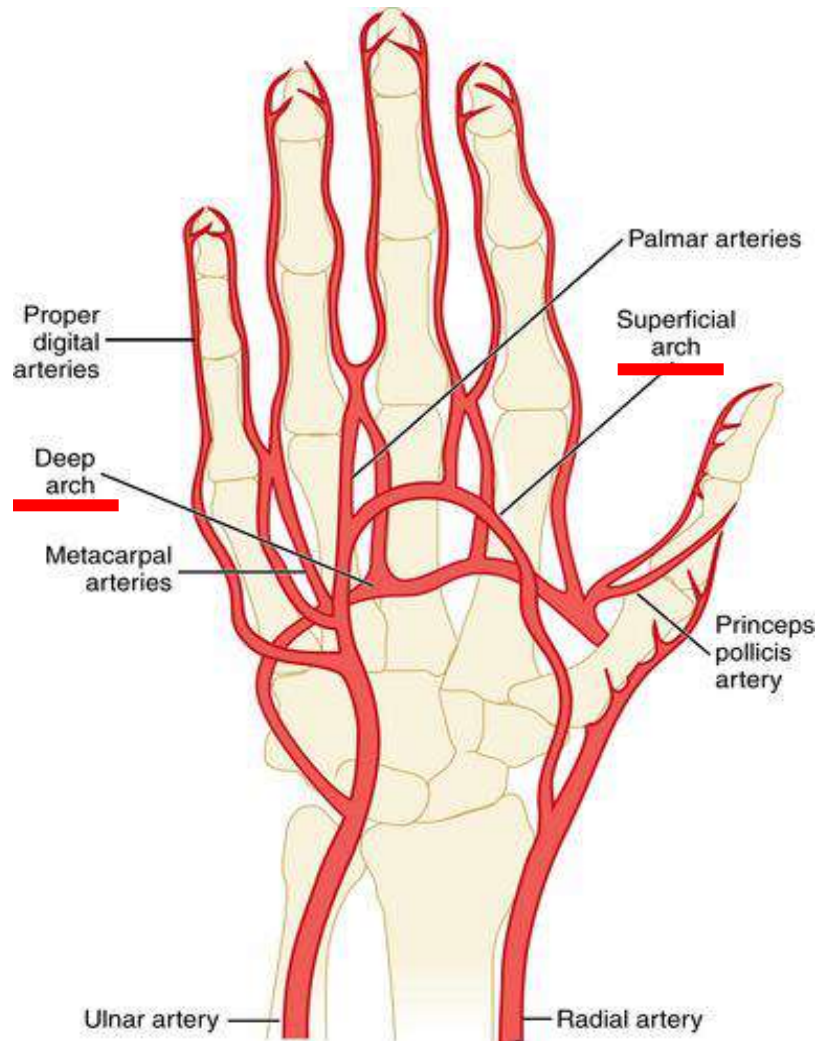


Arteries of the hand

Superficial & Deep palmar arches



Radial & Ulnar Arteries in the hand forming the deep and superficial palmar arches



Ulnar artery

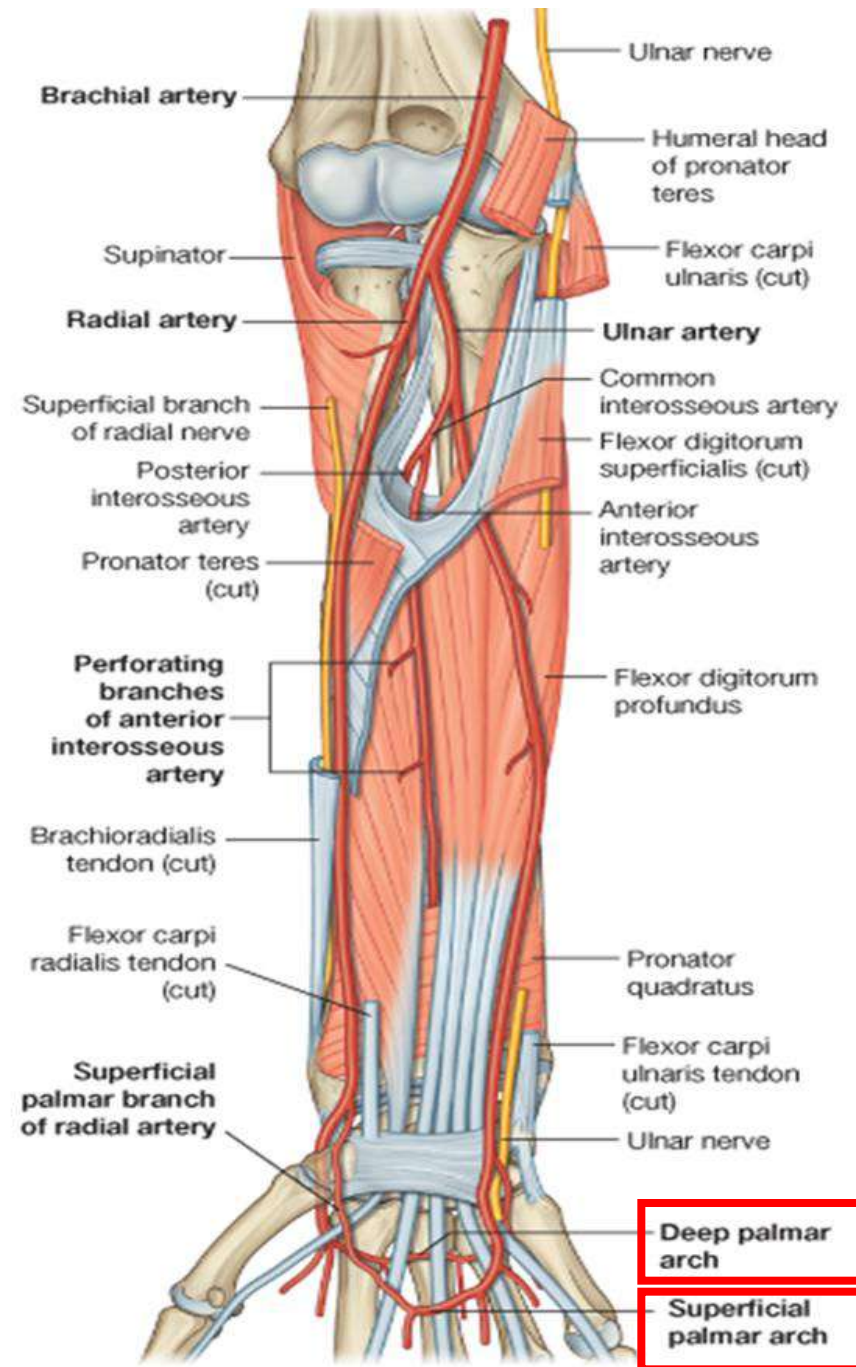
Beginning : it arises as the **larger medial branch** of the brachial artery in the cubital fossa at the level of radial neck.

Ending in the hand : by dividing into **SUPERFICIAL palmar arch & deep palmar branch** .

Radial artery

Beginning : it arises as the smaller lateral branch of the brachial artery in the cubital fossa at the level of radial neck .

Ending in the hand : by dividing into **princeps pollicis artery** & **DEEP palmar arch**



Superficial & Deep Palmar Arches

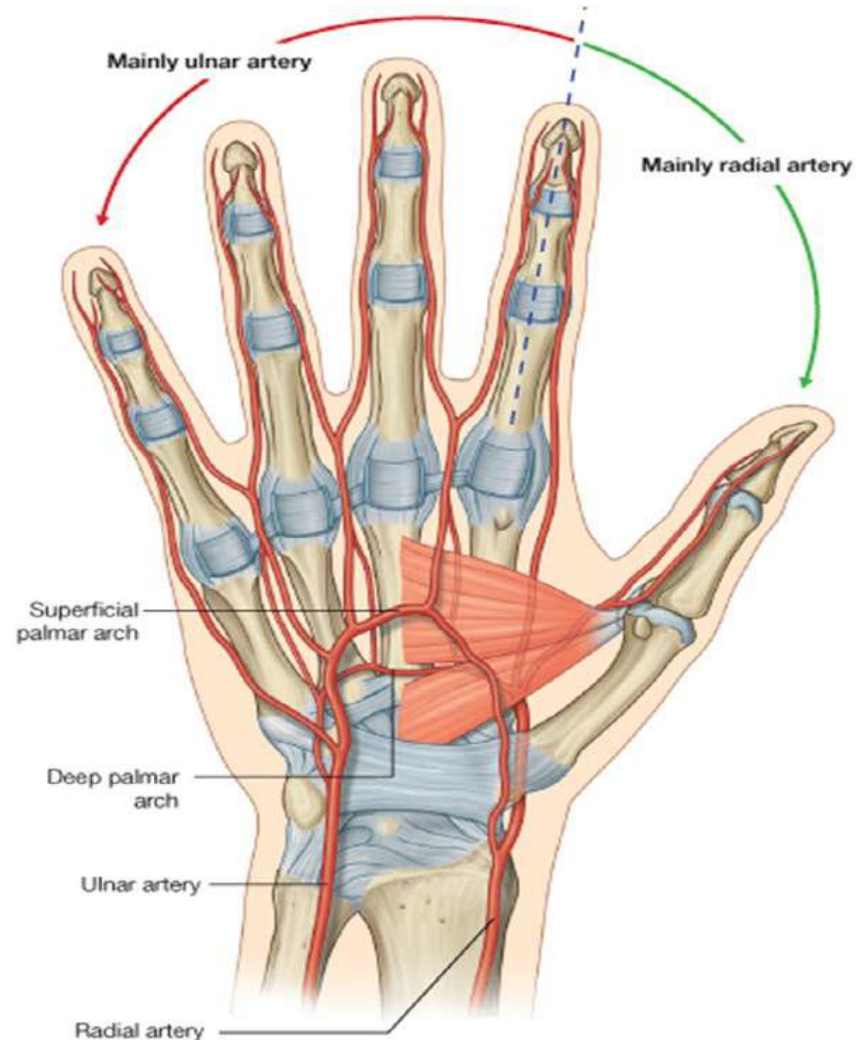
Radial & Ulnar arteries in the Palm of the Hand forming **Superficial & Deep palmar arches**

-Superficial palmar arch :

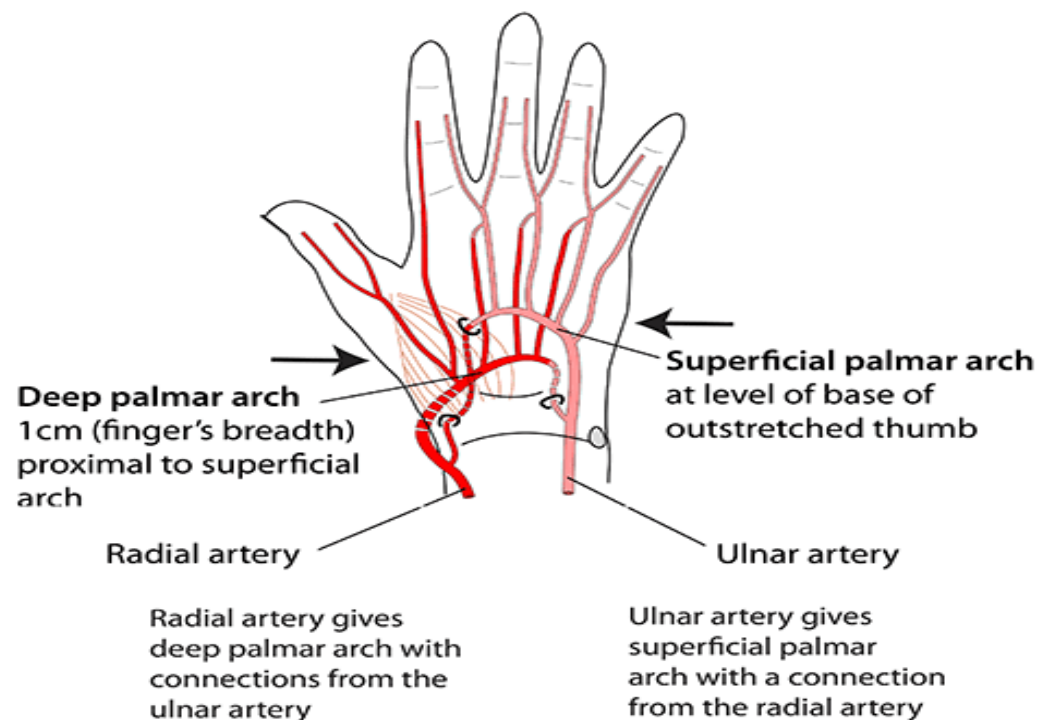
is formed mainly by the **Ulnar artery** and is communicated laterally with the superficial branch of radial artery .

-Deep palmar arch :

is formed mainly by the **Radial artery** and is communicated medially with the deep branch of ulnar artery .



	Superficial Palmar Arch	Deep Palmar Arch
Site	Deep to palmar aponeurosis & Superficial to the long flexor tendons . -At the level of outstretched thumb	Deep to the long flexor tendons. Superficial to the metacarpal bones and interossei muscles. -It lies 1cm proximal to superficial arch
Formation	Is formed mainly by Ulnar artery & is communicated with superficial branch of radial artery	Is formed mainly by radial artery & is communicated with deep branch of ulnar artery
Branches	1. 3 common palmar digital branches 2. Digital branch to medial side of little finger	3 Common carpal metacarpal branches which anastomose with the digital branches of the superficial palmar arch

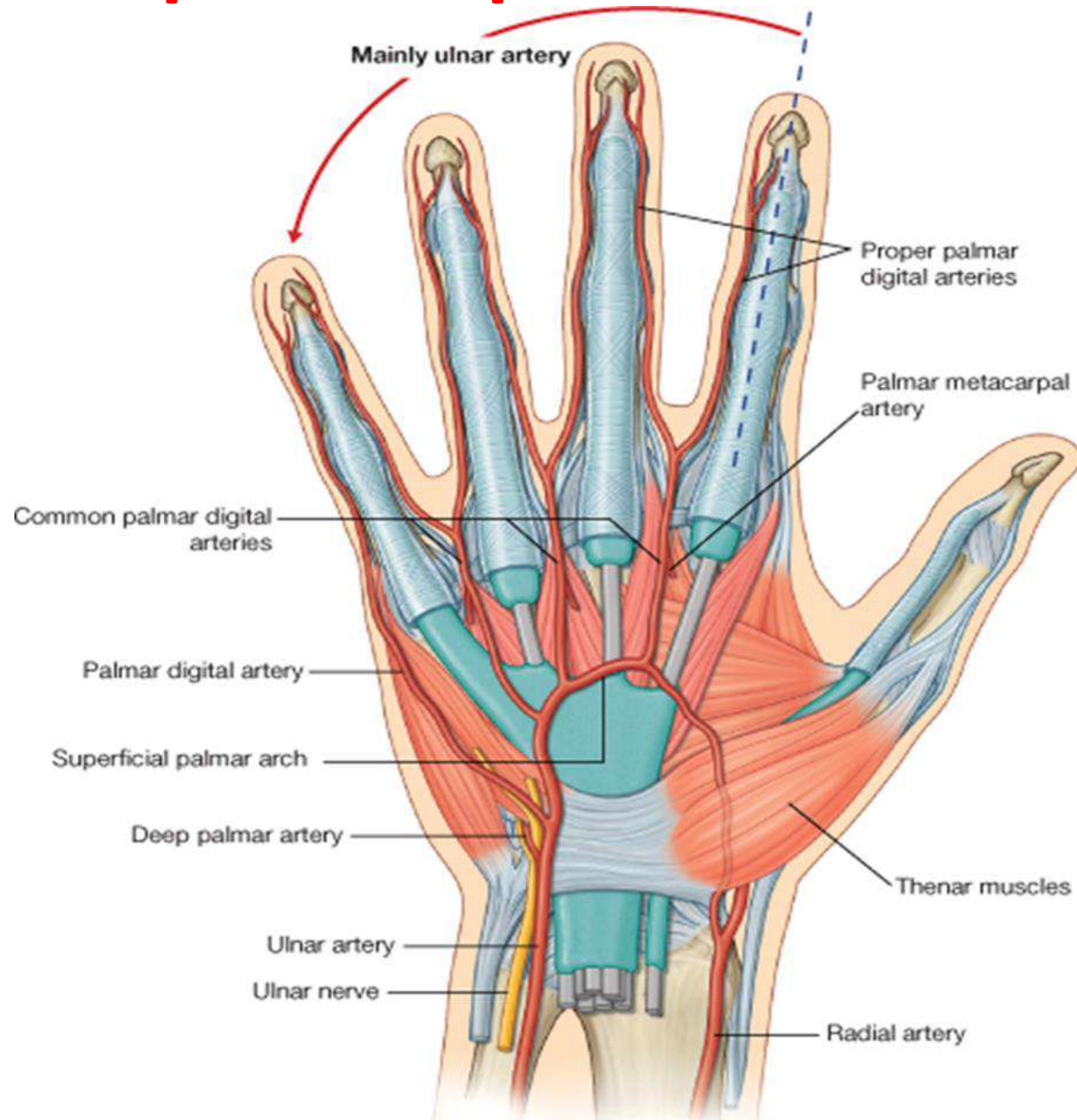


Branches of superficial palmar arch

Branches :

A. 3 common palmar digital branches

B. Digital branch to medial side of little finger .

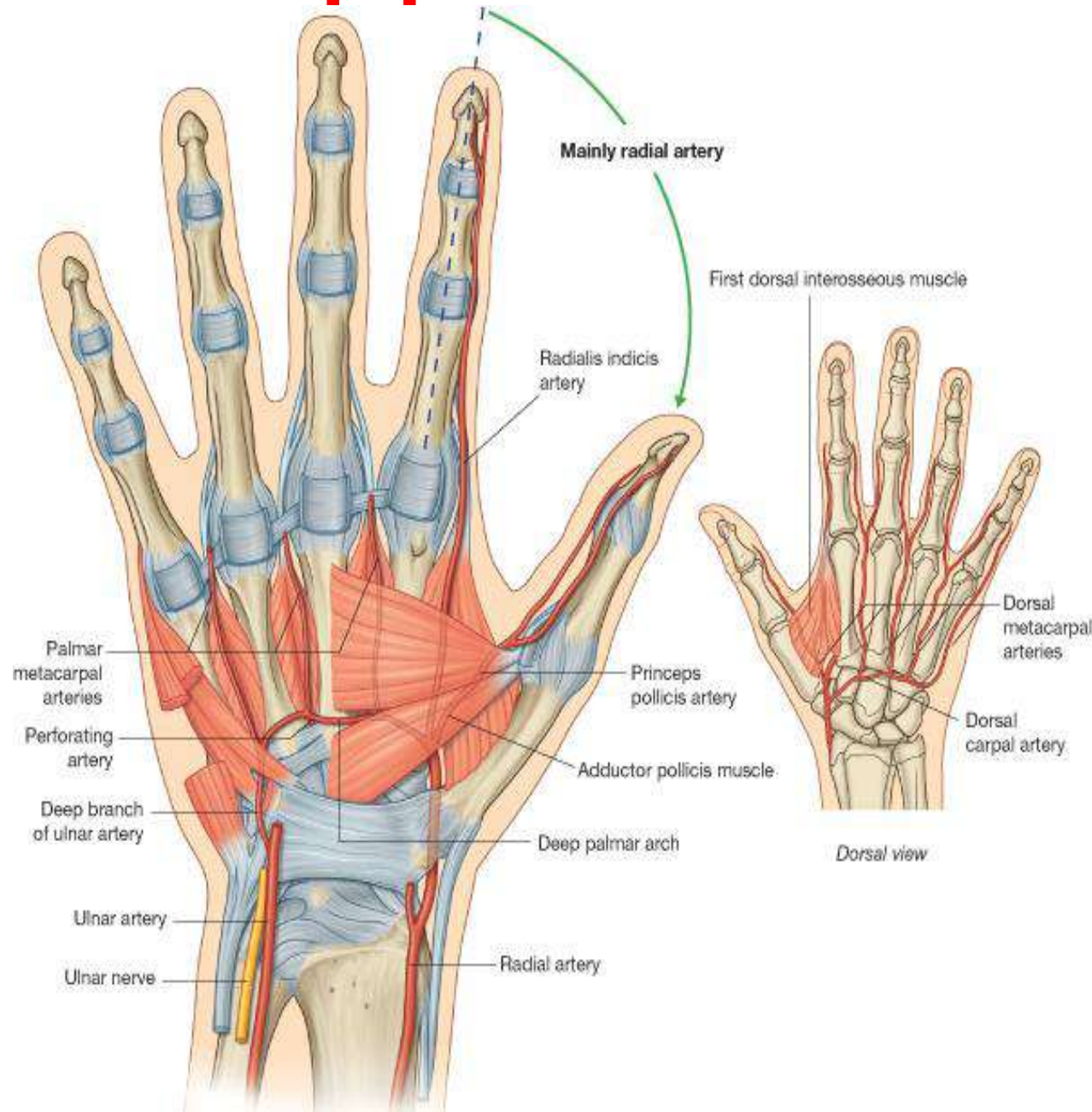


Branches of deep palmar arch

Branches :

A. 3 common palmar metacarpal arteries which anastomose with the digital branches of the superficial palmar arch

B. Recurrent branches :
To join the anterior carpal arch



Anatomical snuffbox

Definition : it is a surface anatomy feature .

*It appears as a triangular depression on the lateral surface of the wrist on full extension of the thumb .

Boundaries :

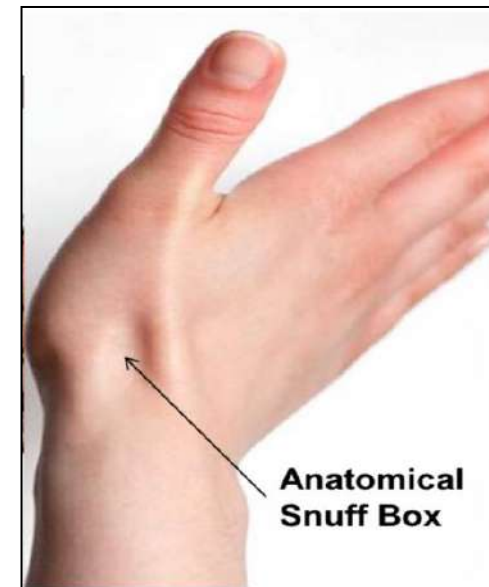
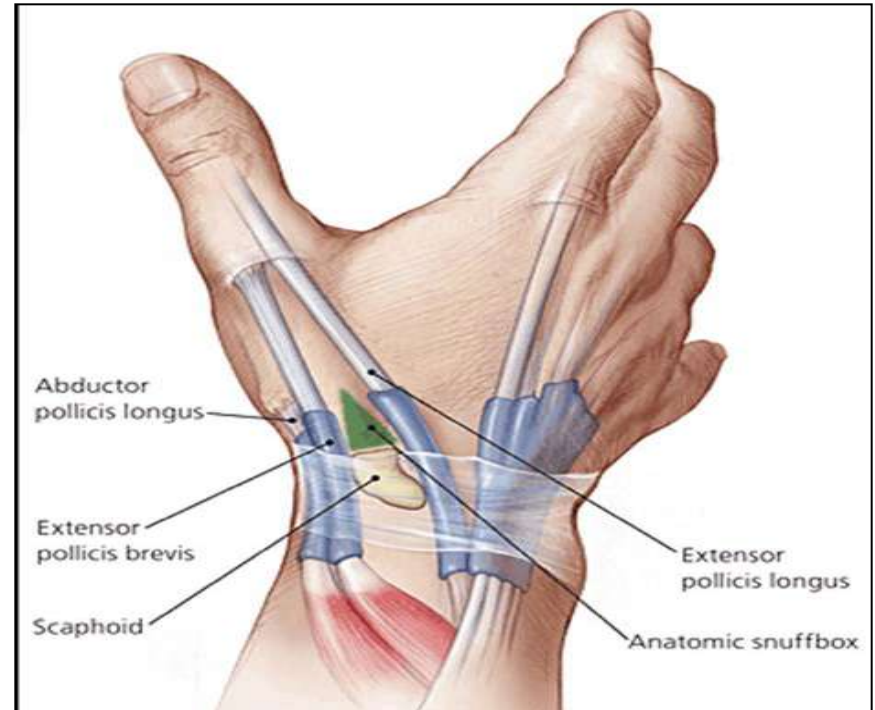
1. **Medial border** : extensor pollicis longus
2. **Lateral border** : abductor pollicis longus & extensor pollicis brevis
3. **Roof** : cephalic vein & superficial branch of radial nerve
4. **Floor** : scaphoid bone

Content : crossed by **radial artery**

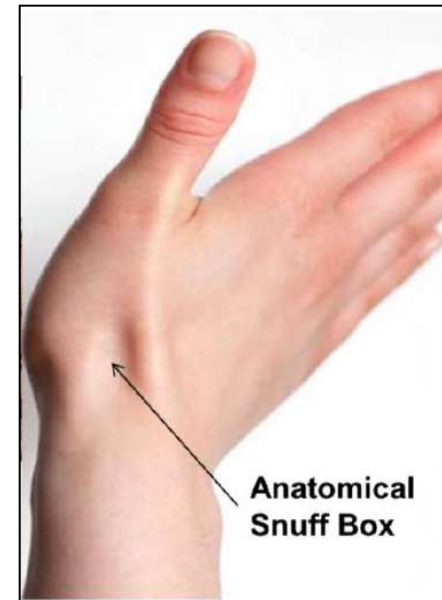
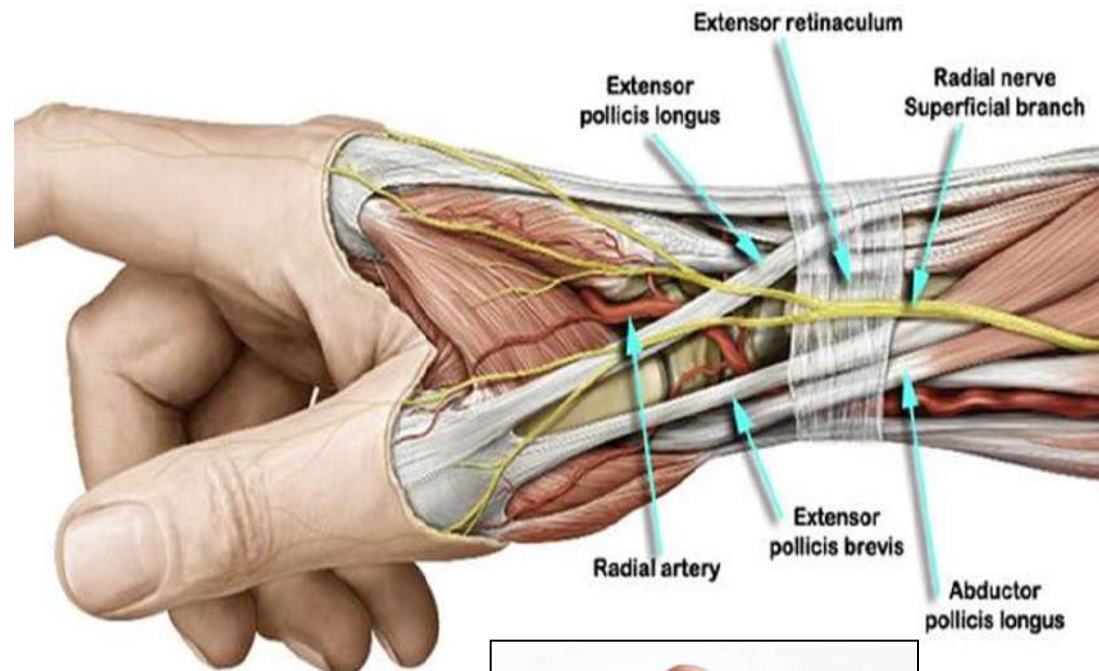
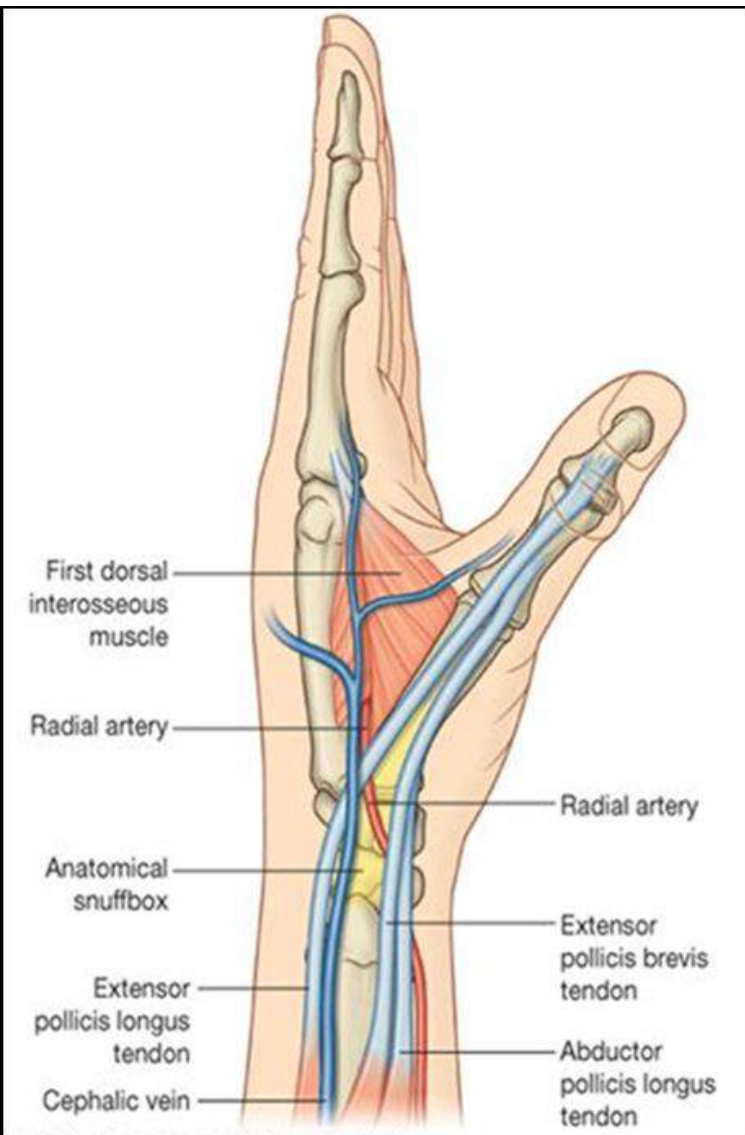
Where its **pulsation** can be felt.

Clinical importance :

1. tenderness in the box signifies **fracture of scaphoid bone**
2. **Radial artery pulsation** can be palpated



Radial artery in the anatomical snuffbox



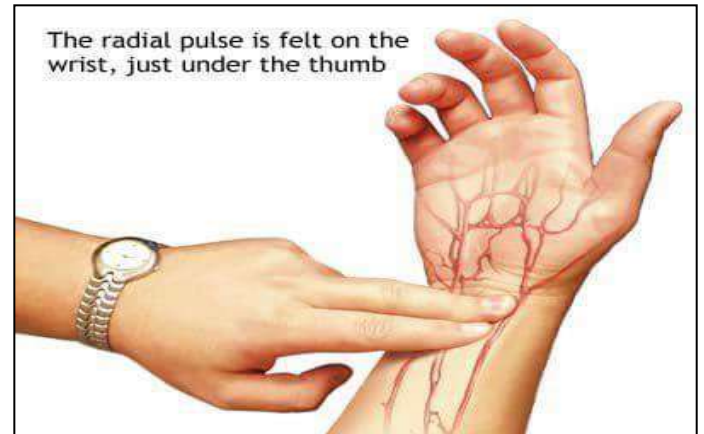
Sites of feeling of arterial pulsations

Brachial artery pulse :

In the cubital fossa just medial to biceps tendon.



Radial artery pulse :
Lateral to the tendon of flexor carpi radialis



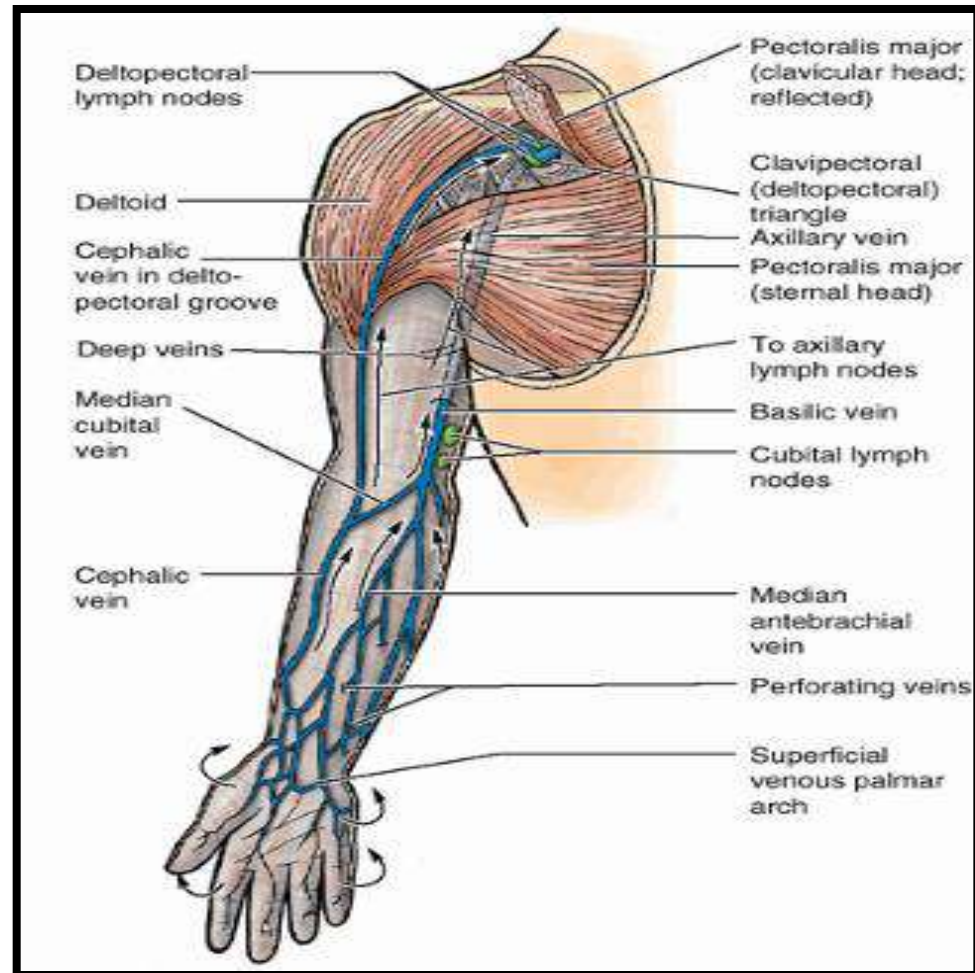
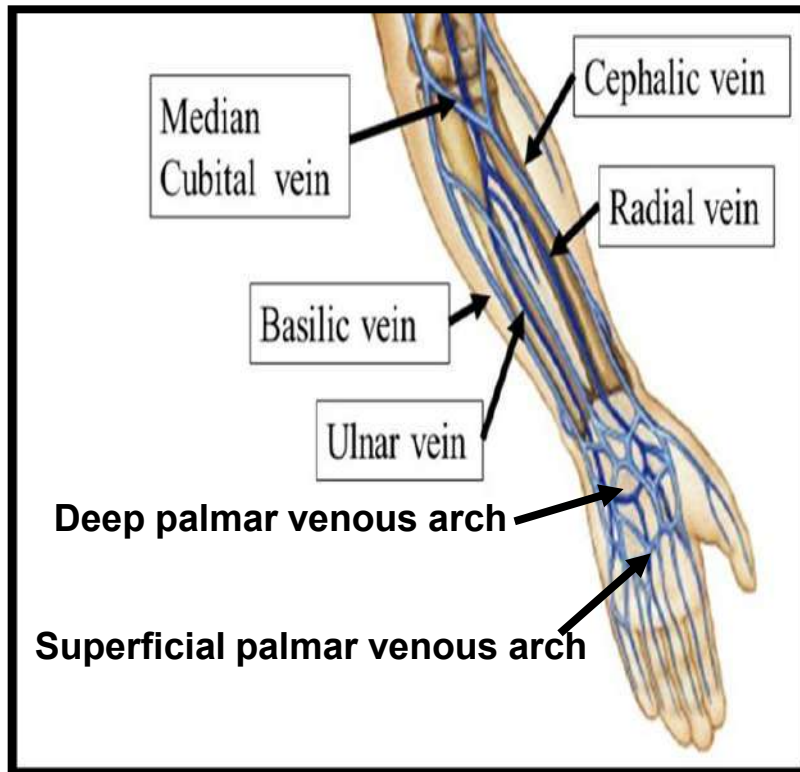
Radial artery in the anatomical snuffbox



Veins of the Palm of the hand

They are **superficial & deep palmar venous arches** .

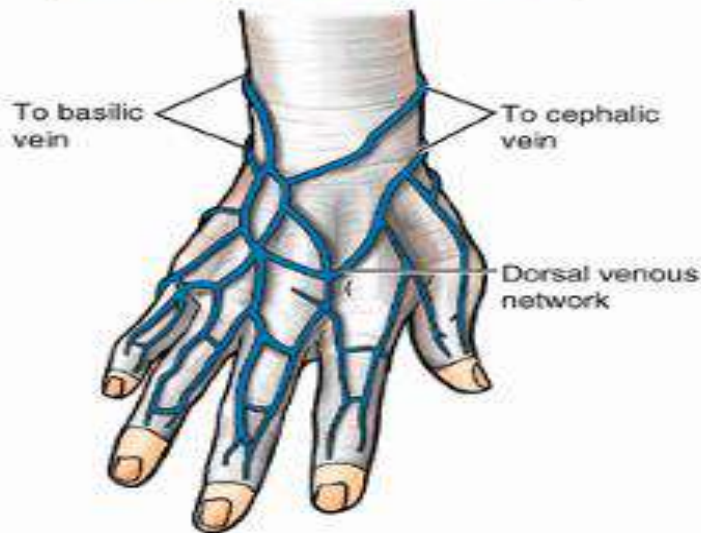
They are accompanied the palmar arterial arches and receiving corresponding tributaries.



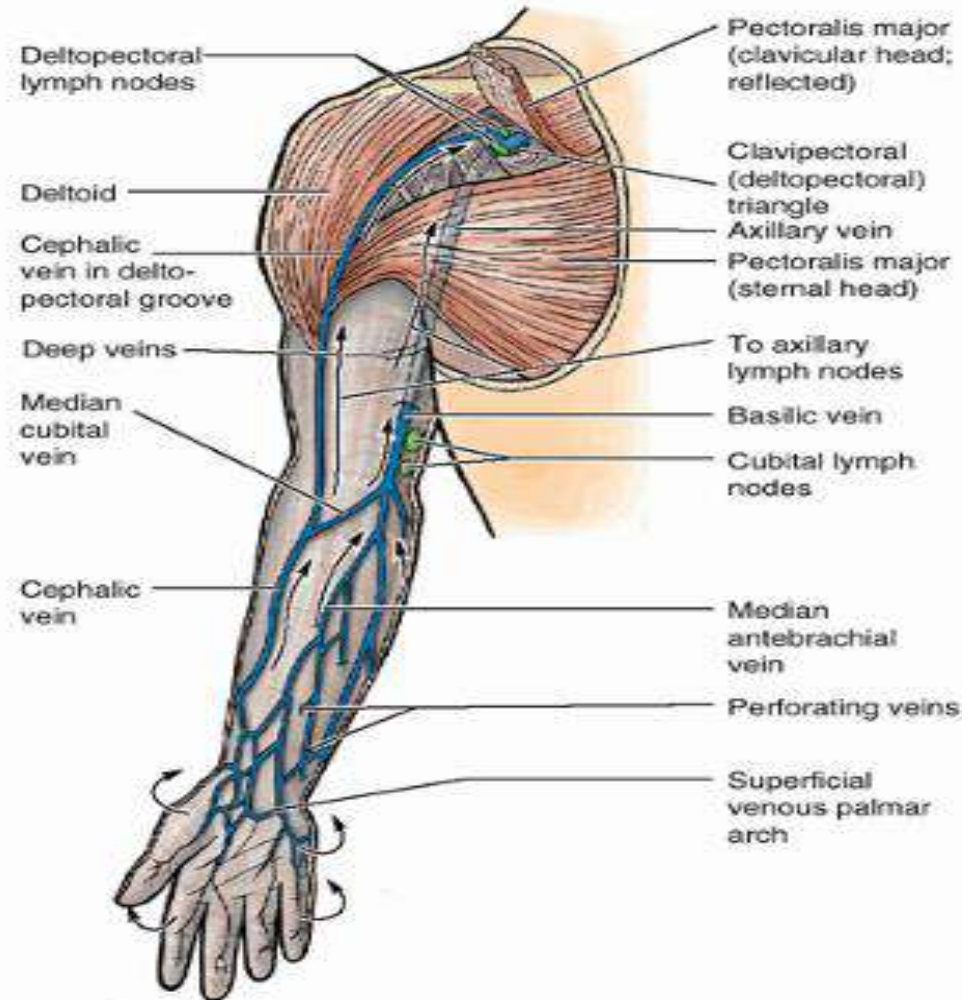
Dorsal venous arch

It is in the form of network on the **dorsum of the hand** .

Its **lateral end** forms the **cephalic vein** while its **medial end** forms the **basilic vein**



(A) Posterior (dorsal) views



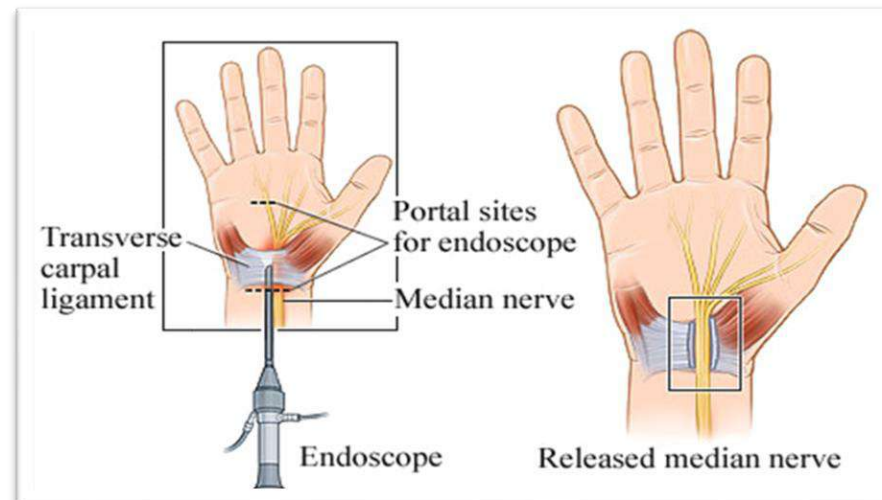
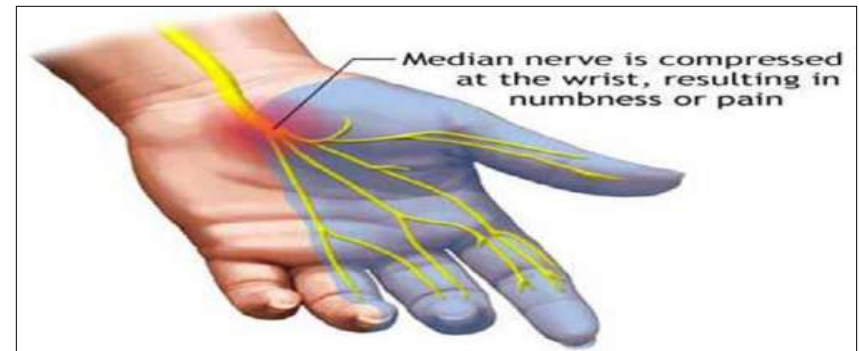
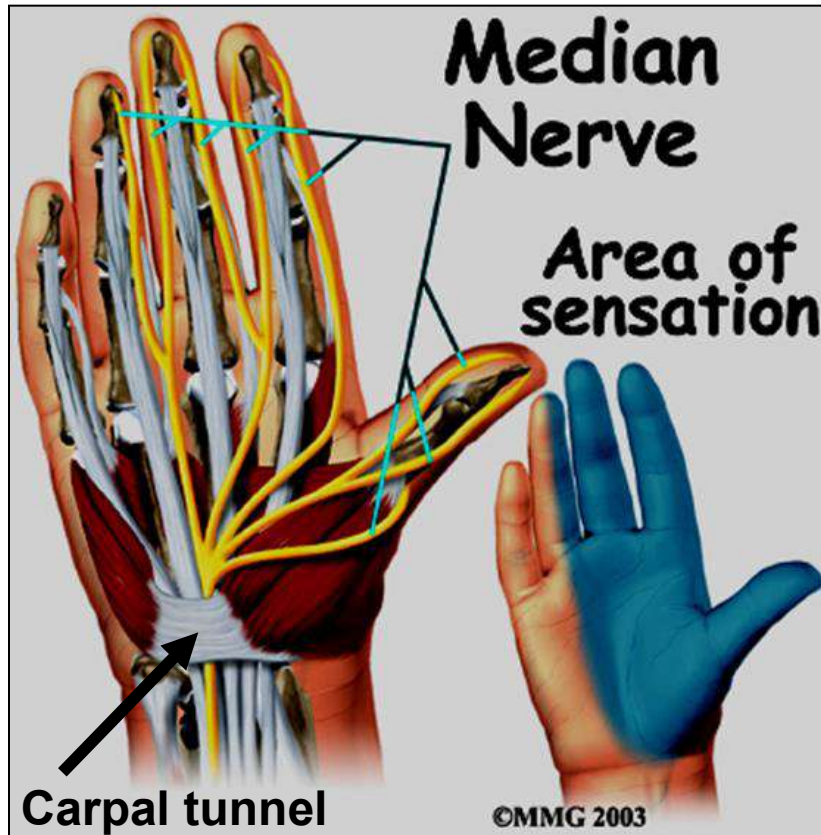
(B) Anterior (palmar) views

Carpal tunnel syndrome

It results from compression of median nerve in the carpal tunnel
(the space between the carpal bones & the flexor retinaculum).

- **Motor effect:** Weakness in abduction and opposition of thumb .
- **Sensory effect:** pain & numbness in the lateral 3 ½ fingers.

No sensory loss in the palm of the hand because the cutaneous branch of median nerve passes superficial to the flexor retinaculum



SYNOVIAL SHEATHS SURROUND FLEXOR TENDONS

COMMON SYNOVIAL SHEATH
FOR FLEXOR DIGITORUM
SUPERFICIALIS AND PROFUNDUS;
EXTENDS UNDER
FLEXOR RETINACULUM

SYNOVIAL SHEATH
FOR FLEXOR POLLICIS
LONGUS

DIGITAL
SYNOVIAL SHEATH
IN FINGERS

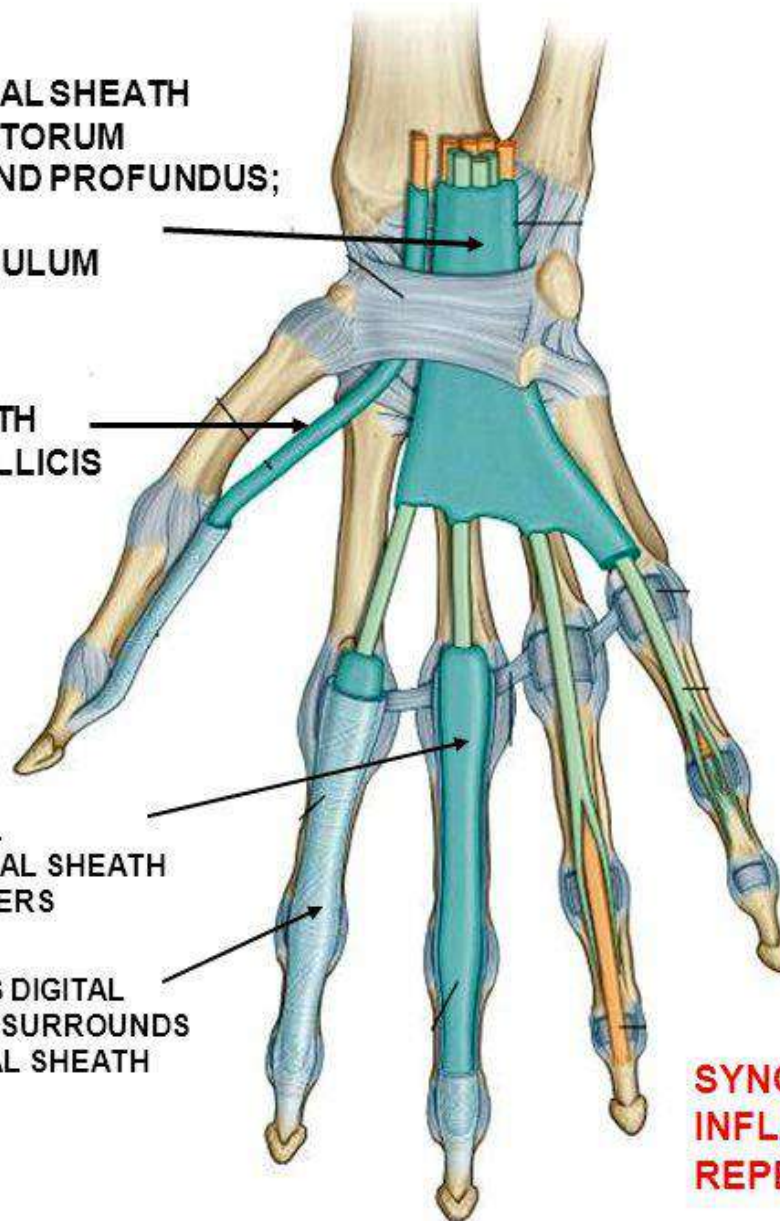
FIBROUS DIGITAL
SHEATH SURROUNDS
SYNOVIAL SHEATH

- SYNOVIAL
SHEATHS
SECRETE
SYNOVIAL FLUID

- ACTS AS
LUBRICANT

- REDUCES
FRICTION WHEN
TENDONS
MOVE UNDER
FLEXOR
RETINACULUM

**SYNOVIAL SHEATHS CAN BECOME
INFLAMED AND SWOLLEN WITH
REPETITIVE MOVEMENTS**



Tenosynovitis

Infection of the tendon with its synovial sheath

- **Closed space infection of flexor tendon sheath of digit.**
- **Results from penetrating trauma or hematogenous spreading.**

Flexor Tenosynovitis

- Finger in slight flexion
- Fusiform swelling
- Pain with extension.
- Tenderness along tendon sheath

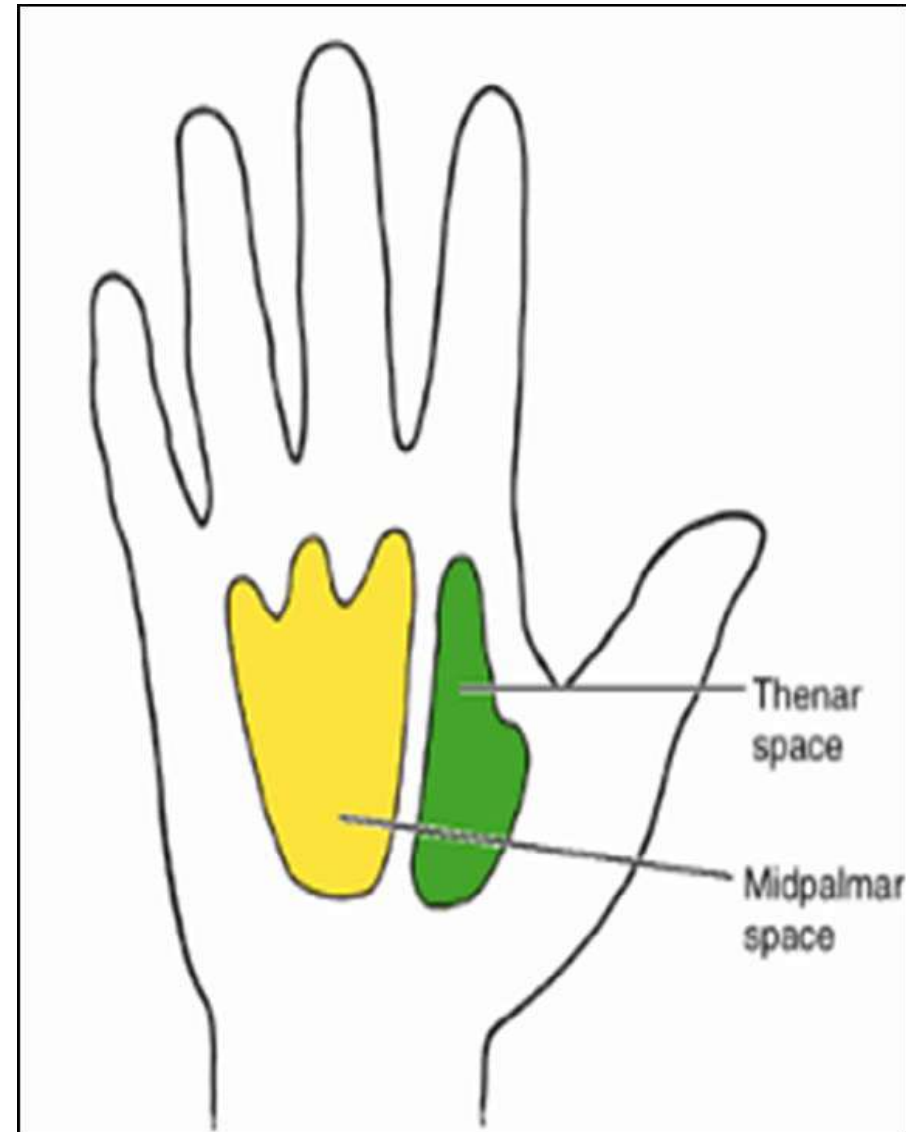


Fascial spaces of the palm

They are fascial spaces deep to the palmar aponeurosis and divided by a **midpalmar (oblique) septum** into : **thenar space** & **midpalmar space**

- The thenar space contains the first lumbrical muscle and lies posterior to the long flexor tendons to the index finger and in front of the adductor pollicis muscle
- The midpalmar space contains the second, third, and fourth lumbrical muscles and lies posterior to the long flexor tendons to the middle, ring, and little fingers

They are important clinically for containing infections



الحمد لله
على ما تعاقب الليل والنهار

Extensor Retinaculum

Remember!

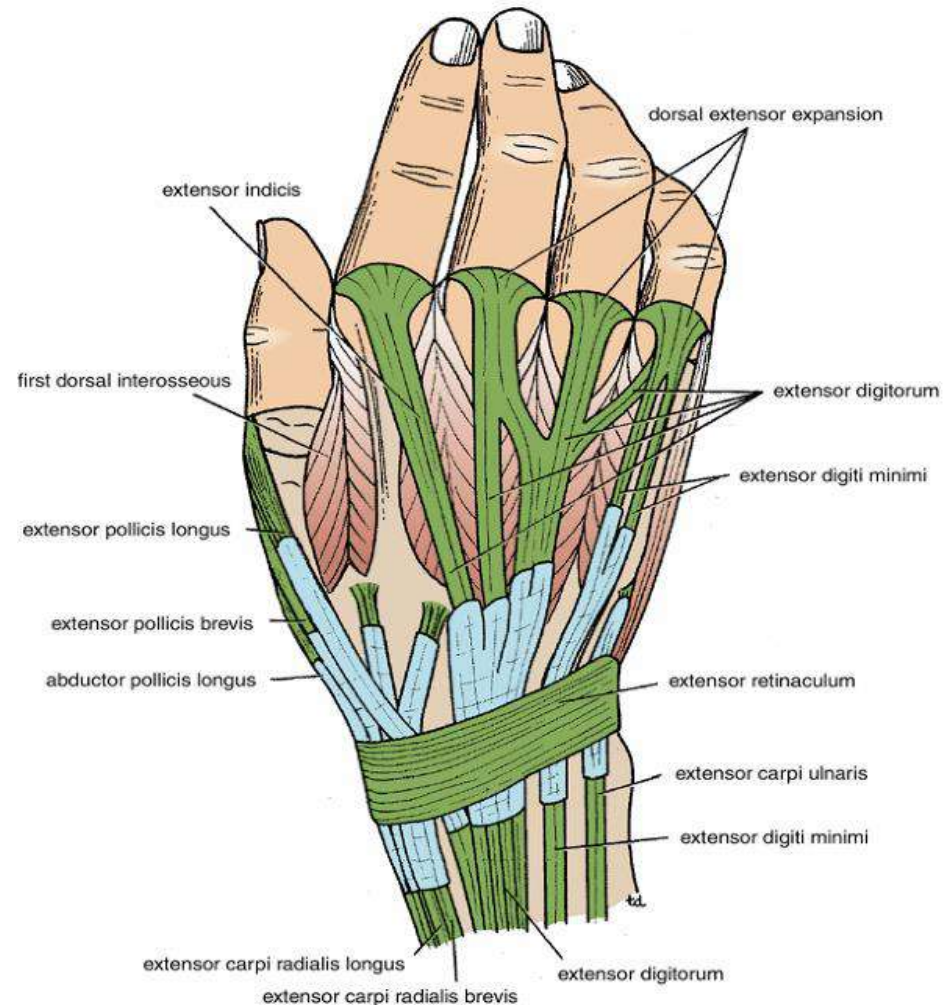
Attachment of extensor retinaculum

1- Medial end : to triquetrum

& styloid process of ulna

2-Lateral end : lower part of
anterior surface of radius

3- Inferiorly : it sends 5 septa
attached to the ridges on
posterior surface of lower
end of radius & ulna,
dividing the space under it
into 6 compartments .



All the muscle of posterior compartment passing deep to the Extensor Retinaculum (in 6 compartments)

Remember!

6 compartments from lateral to medial :

**1- a-Abductor pollicis longus
b-Extensor pollicis brevis**

**2- a-Extensor carpi radialis longus
b- Extensor carpi radialis brevis**

3- Extensor pollicis longus

**4- a- Extensor digitorum
b- Extensor indicis**

**c- Terminal part of posterior interosseous nerve
d- Terminal part of posterior interosseous artery**

**5- Extensor digit minimi
6- Extensor carpi ulnaris**

